Manuscript outline: what each paragraph should contain

enceladus_DSM `studies/keff_sintering/manuscript_outline_new.tex`

Re-synced 2026-10-15 against the Overleaf project uploaded that day (second upload), with the literature check merged in

Literature check of the 10-15 draft: ranked index

Three independent reviews (`Literature/review/{A,B,C}`), merged and re-checked page by page against the library. The evidence (paper, PDF page, short quote) and any *suggested text* sit in the body, at the entries marked **[lit. check Fn]** . **F1–F5** change what the paper claims or how it justifies a choice; settle them first.

F1 **The gap must be stated against Bouvet et al. (2022)**, who computed k_{eff} through isothermal metamorphism; the gradient sentence is Calonne et al. (2014), not 2011. (Intro ¶8.)

F2 $\alpha_c = 10^{-3}$ **is attachment-limited, not diffusion-limited.** (Intro ¶6; Section 3.2.)

F3 **Surface diffusion does not explain the** $-20\,^{\circ}$**C shortfall**, and the model gives 41% *of* the growth. (Section 5.1.)

F4 **Exponent sources**: Hobbs & Mason give 1/5 for vapor through air; Thomas et al. give 4–5 for surface diffusion; the exponent alone does not identify the mechanism; Kingery's 1/7 is disputed; "planetary timescales" reverses Molaro. (Intro ¶5; Section 5.1.)

F5 **The 2 mm domain cannot rest on Calonne et al. (2011)**, whose REVs are 2.5–5.5 mm; use our domain-size test. (Section 2.5, Aggregates.)

F6 **New ¶4**: "at least 180 K"; "strengthening"; the 0.01–1 mm yr^{-1} range is Choukroun's summary; burial and conductivity sentences need citations. (Intro ¶4.)

F7 **Bouvet et al. (2023) → (2022)** for the time rescaling through α_c. (Section 5.1, α_c as an effective parameter.)

F8 α_c **range and form**: cite Kaempfer & Plapp and Braun et al.; not Arrhenius. (Intro ¶6.)

F9 **Equations**: P_a vs P_{atm}; a stray T_0; k_j source; "ASHRAE"; empty scaling appendix (`Sclaing`); Eq. (11d) missing ∇. (Sections 2.1, 2.3.)

F10 **Smaller citation fixes**: Colbeck (1998) for depth hoar; Postberg's 6%; Swinkels & Ashby; Colbeck (1983) for experiments; Flin for bulk properties; Auriault et al. (2009) book; Nicoli's scalar law. (Intro ¶2, ¶3, ¶5, ¶7, ¶8; Section 2.)

F11 **Bibliography**: Byrne et al. (2024) is the preprint (use doi:10.1029/2024JE008615); `noauthor_sintering_nodate` duplicates `Hobbs_and_Mason`; Molaro pages 243–276; Calonne (2011) L23501; Wexler pages 5–20; Flin "2014" is *Physics and Chemistry of Ice 2010*, Hokkaido Univ. Press, 2011, pp. 321–328.

F12 **Typos** in the new text are listed under each paragraph.

Citations worth adding (closely matching, checked): Legagneux & Dominé (2005, p.9), whose vapor-transport model shows temperature as "a translation of the curve along the time axis" (for Section 5.2); Calonne et al. (2019, p.5), bonds "act as heat paths" at low density (for Section 5.3). **Not checkable here:** Hobbs & Mason (1964), Massman (1998), Moure & Fu's Supporting Information.

Framing. Dry snow metamorphism is the story; the model covers one part of it, sintering by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach. Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to $-5\,°C$, air at 1 atm, sealed pores, 50 μm grains, 30 days, two dimensions.

How to read this. Every paragraph of the draft has a numbered entry: its subject, where it is, a one-line overview, and bullets for what I suggest changing. Line numbers (L first–last) are those of the source of 2026-10-14 compiled here (31 pages; the body ends at L492). That upload changed only the Introduction, so the Introduction entries are new from a second full read, and every later entry is the 10-13 review with its line numbers moved up by 16. Each bullet is tagged: **[correct]** a statement that is wrong or unsupported; **[wording]** grammar or wording; **[trim]** text that can go or shrink; **[add]** something missing. A paragraph with no **[correct]** bullet says nothing I found to be wrong. In the Introduction, **[add]** bullets marked **[from the notes]** or "(discussed 10-xx)" are points we have already worked out or that the old Discussion carried, and which the draft does not yet contain. Figure captions are reviewed in their own section, after Section 6. Entries marked "to write" do not exist yet; their bullets are the content. Boxes mark where a figure or table goes. From Section 5.2, paragraph 5, on, the entries keep the earlier form (a "Message" and a content list), since nothing is drafted there.

First round (user, 2026-10-14). This round checks that what the draft says is correct, free of typos, not missing a major argument, well ordered and concise. A paragraph that does all that is marked "no changes"; fine polish waits for the second round, after the team has read it.

The old Discussion is gone from the source. Its literature points are kept here, marked **[from the notes]** . Their line numbers refer to the PDF of 2026-10-12 in `Literature/`, which still contains that text. Each point is to be checked against its source before use.

Suggested citations follow each paragraph, with the key in your `.bib`. Most are the ones the old Discussion used. I have matched them by title only; check each claim against the paper before it goes in.

Decisions already made, which this plan follows

Collected from the work of 2026-09-08 to 2026-10-12: the session log, the campaign record and the manuscript plan.

- **Framing** (10-08, 10-10). Laboratory-observable sintering by vapor transport, inside the dry-snow-metamorphism story. The Introduction and title stay as written; scope caveats go in the Methods. No timescales for Enceladus temperatures; the timescale map is dropped. Advisor's comments still to come.
- **Comparisons** (10-08). Not natural snow (angular, more porous, loaded). The analogues are unloaded ice-sphere and ice-powder samples: Molaro et al., Demmenie et al., Choukroun et al. From the snow literature only functional forms carry over, never magnitudes.

- **Kinetics** (09-22, 10-09). α_c constant in temperature; the thin-interface term always on; aggregates stated as $\alpha_c = 1.020 \times 10^{-3}$. No further simulations for this paper.
- **Two-grain argument** (09-09, 10-02, 10-13). Rests on what the vapor route does and does not reproduce, and on exponents; no Thomas or Kingery runs. The two mismatches have different causes and neither is attributed to missing surface diffusion (Appendix, this outline): the rate at $-20\,^{\circ}\mathrm{C}$ is a matter of how fast vapor can be supplied, in an experiment whose driving force is a few millikelvin-equivalent; the exponent at saturation now matches: from grains just touching the model gives 0.29, inside Demmenie's 0.26–0.33 and on the sharp-interface law (10-15); the earlier 0.21 came from a pre-made starting neck. Molaro's conventions: neck *width*, time anchored on the first measurement. The $-20\,^{\circ}\mathrm{C}$ series is the benchmark.
- **Conductivity** (09-23). Tensor law in the interface; the cell problem is a steady problem on a frozen microstructure and says nothing about the transient model. Two dimensions only, so no absolute k_{eff} is claimed.
- **Clock and normalization** (10-08, 10-12). The clock is τ_{sub}; curves are normalized at $\theta_r = 30$. The first day depends on the interface parameter, so a "rise from day 1" is never quoted as a property of the ice.
- **Notation** (09-28 to 09-30). ϕ phase field, φ porosity, ζ indicator, χ curvature; subscripts "ice" and "air", not i and a; no index notation; a single dot for every contraction; the vector $\mathbf{t}$ named as in Auriault et al.
- **Citations** (09-23). Only work whose setting matches ours; every DOI checked against Crossref.
- **Words.** Desublimation ("deposition" is plume fallout); "vapor"; ε is the interface parameter, not the interface width.

Where the draft stands

Section	State
Key points, Abstract, Summary	to write (placeholders)
1 Introduction	third draft (10-15): new ¶4 on Enceladus conductivity; open: the gap against Bouvet et al. (2022), the exponent sources, α_c regime, why SSA
2.1–2.3 Model, kinetics, homogenization	written; T **back to °C (10-16)**, $T + T_0$ where kelvin is needed; scaling appendix empty; Eqs. (2), (11d) to fix
2.4, 2.5 Numerics, simulation design	headings only
3 Grain pairs	written; 3.2 to rewrite for the tangent-start run ($a = 0.29$, Figure 4 rebuilt 10-15)
4 Aggregates	written; numbers to add, one paragraph to correct
5.1 What vapor transport reproduces	three paragraphs drafted; the first two need revising for what was found on 10-13
5.2 Temperature as a rescaling of time	one paragraph begun
5.3–5.6, 6 Conclusion	headings only
Appendices A, B	headings only; content is in the Appendix of this outline

Done since the 10-14 upload (the 10-15 drafts): new Introduction ¶4 (Enceladus deposits and conductivity); ¶5 rebuilt around the exponents with Hobbs & Mason, Kingery, Thomas and Demmenie in the `.bib`; ¶6 with Libbrecht & Rickerby, layer nucleation and a constant α_c; ¶7 with Braun et al. described; ¶8 ending on the paper's gap; ¶9's two findings restated; Figure 1 caption vapor-only; kelvin throughout Section 2.1 (reversed to °C on 10-16) with Eq. (3) in Wexler's form; ρ_i in the latent term; "Allen-Cahn"; "Modeling the"; the comma in Ahmed Mahjoub's name; `finalold` class. The entries below list only what is still open.

Work list, in order

1. **Correct Section 5.1 as drafted** (L426–477). Two sentences attribute the shortfall to surface diffusion, which Appendix A rules out; "by 41% and 170%" misstates the numbers; one sentence is unfinished. Details under Section 5.1 below.
2. **Restate the gap against Bouvet et al. (2022)** (Intro ¶8, F1) and fix the exponent sources (¶5, F4).
3. **Correct the α_c regime** in Intro ¶6 ("diffusion limited") and Section 3.2 (tex L455, "vapor-limited"): $\alpha_c = 10^{-3}$ is attachment-limited (F2).
4. **Rewrite Sections 3.2, 5.1 and Appendix B for the tangent-start run** (plans and suggested text under each), now in Figure 4: the draft text (tex L455–461, L502) still describes the pre-necked run (a 28 μm starting neck, exponent 0.21). **[written] Tangent-start result (10-15)**: free exponent $a = 0.289$ (Kuczynski $1/m = 0.290$; 0.277 from later window starts), inside Demmenie's 0.26–0.33 and on the sharp-interface 0.29; no trapped pocket; neck to 61.2 μm in 150 h. The pre-necked 0.21 came from the initial condition. Figure 4 is rebuilt from this run (`studies/molaro_2019/demmenie/README.md`).
5. **Write Sections 2.4 and 2.5**; move the set-up text out of Sections 3 and 4 into them.
6. **Add the numbers to Section 4** and correct the high-porosity paragraph (L382–400).
7. **Finish 5.2; write 5.3–5.6, the Conclusion, Appendices A and B.** Plans below; the two-page scaling note (`temperature_scaling/`) is the source for 5.2.
8. **Front matter:** key points, abstract, plain-language summary.
9. **Small corrections** (last section of this outline).

Runs pending

- **Saturated pair from tangent contact**: done (job 4474214). **[written] Tangent-start result (10-15)**: free exponent $a = 0.289$ (Kuczynski $1/m = 0.290$; 0.277 from later window starts), inside Demmenie's 0.26–0.33 and on the sharp-interface 0.29; no trapped pocket; neck to 61.2 μm in 150 h. The pre-necked 0.21 came from the initial condition. Figure 4 is rebuilt from this run (`studies/molaro_2019/demmenie/README.md`).
- **Molaro $-20\,^{\circ}$C from tangent contact** (prepared, not submitted). A 10-minute pilot (batch file `molaro_tangent_T-20_pilot`) checks first for a trapped void at the neck, the artifact that led to the pre-made neck in August. **[written]** Pilot run 10-15: the enclosed air pocket forms again (at $r = 3.3$ μm by 60 s), at half the interface parameter. Not submitted.

- The passages marked "provisional" below can now be written from the tangent-start run.

Section titles

The descriptive headings are adopted in the draft. "2 Modelling Approach": AGU uses American spelling, and the draft writes "modeling" elsewhere, so "Modeling". Still to change: "2.2 Gibbs-Thomson Condition" (Interface Kinetics and the Sintering Time Scale), "2.3 Homogenizing for Effective Thermal Conductivity" (Effective Thermal Conductivity by Homogenization), and capitals in "5.1 What vapor transport alone reproduces". All optional.

Build check (LaTeX warnings)

The project uploaded on 2026-10-15 (second upload, `finalold` class) compiles here in 25 pages with **no warnings** other than the one below, and BibTeX reports none. What remains:

1. **"Label(s) may have changed. Rerun"** appears after any edit that moves a label and goes away on the next compile. Overleaf reruns automatically; if it persists there, recompile from scratch (the "Recompile" menu, "Clear cached files").
2. **Not cited in the text** (no warning, but AGU requires it): Table 2 (temperature-dependent quantities), Table 3 (complete parameters), Figure 2.
3. **Unused files** in the project: the Figure 8 timescale map, the all-porosity variant of Figure 7, and the AGU template file.

If Overleaf shows warnings other than these, paste them: its compiler settings can differ from the build here.

Front matter

1. **Title and authors** (L1–4)

   The paper's name and author list.

   - **[written]** "Modeling the"; "Ahmed Mahjoub".
   - **[wording]** Remove "–Version 2" from the title.

2. **Key points** (L7–8, placeholder)

   Three statements of at most 140 characters each.

   - **[add]** For a given ice-grain packing, temperature from -40 to $-5\,^{\circ}\mathrm{C}$ only rescales the time over which sintering raises conductivity.
   - **[add]** Effective conductivity follows specific surface area by one power law in well-connected packings; porosity sets its level.
   - **[add]** At saturation, vapor transport alone gives the neck-growth exponent measured between ice spheres (0.29 against 0.26–0.33).

3. **Abstract** (L9–31, marked "needs to be updated")

   Background, method, the two-grain test, the aggregate result.

- [correct] "Driven by temperature gradients": every run here is isothermal.
- [correct] "replicating experimental results for various temperatures without retuning" and "incorporating a temperature dependence into our kinetic parameters measured in experiments": neither is done; α_c is constant and the model gives 41% of the measured growth.
- [correct] "collapse ... for both varying temperatures and varying porosities": across temperature yes; across porosity only the exponent is shared, for $\varphi \leq 0.375$.
- [trim] The first five sentences are background; keep two.
- [add] Sentence by sentence: setting; gap (conductivity as grains sinter without compaction); method; grain pairs (41% at $-20\,^{\circ}\mathrm{C}$; exponent 0.29 at saturation, within the measured 0.26–0.33); aggregates (125 runs); result 1 (same conductivity against SSA at every temperature within 0.3%); result 2 ($\mathrm{SSA}^{-0.80}$ for $\varphi \leq 0.375$, level set by porosity); meaning. At most 250 words (it is 254 now).

4. **Plain Language Summary** (L32–55, marked "needs to be updated")

   The same story for a general reader.

   - [correct] "tune our model at $-20\,^{\circ}\mathrm{C}$ ... without retuning ... strong agreement" and "regardless of temperature or initial porosity": as for the abstract.
   - [correct] "changes in temperatures (sunrise or sunset)" and "gradients in temperature": not part of this study.
   - [wording] "know as" (known); "effecting", "effects" twice, "effect" (affecting, affects, affect); "C.Then" (space).
   - [trim] About 300 words; the limit is 200.

1 Introduction

Re-synced 2026-10-15 against the current draft (upload of 10-15, source lines "tex L"). Nine paragraphs. Each entry lists what the 10-15 draft already did ([written]), then only what is still open, with the literature-check findings (Fn, top of this document) merged in place.

1. **1. What dry snow metamorphism is** (tex L254)

   Defines the process and says it changes connectivity and bulk properties.

   - No changes. Porosity stays in the list (metamorphism in general changes it; the model section says ours does not); the driving-force sentence is right as written; freeze/melt stays.

2. **2. Why it matters on Earth** (tex L257)

   Everyday examples, avalanches, and heat flow through snowpacks and firn.

   - [written] "DSM is why"; the permafrost clause is gone; the conductivity sentence now cites Calonne et al. (2011, 2019) for heat flow and the temperature profile (supported: Calonne et al. 2019, p.2).
   - [correct] [lit. check F10] Depth-hoar sentence: Colbeck (1998) is a theory of bond growth (p.2: "The basic shape of bonds in snow is dictated by the geometrical requirements of grain-boundary grooves"), not depth hoar or avalanches; drop it and keep Fukuzawa & Akitaya and Schweizer et al. "drives slab avalanche initiation" → "forms a weak layer on which slab avalanches can release".

- [correct] The Calonne citation is written `\citet` mid-sentence; it must be `\citep`.
- [wording] "glacial systems": metamorphism acts in snow and firn, not in glacier ice; "snowpacks and firn". "viability of ice in our alpine and polar regions": "persistence of ice in alpine and polar regions". "understanding for how": "understanding of how".
- [add] (optional) Conductivity is also an input of detailed snowpack models, which set it from density alone (Vionnet et al. 2012, p.9, after Yen 1981). This sets up why a microstructural law matters.

3. **3. Icy bodies and the Enceladus plume** (tex L260)

   Water ice in the solar system, ocean worlds, the plume and what its grains are made of.

   - [written] The suggested wording is in ("Many bodies . . . ", *Cassini* italic, "some of which is deposited back").
   - [correct] [lit. check F10] "contain percent level NaCl": only a minority of grains do. Postberg et al. (2009, p.1): "About 6% of these spectra . . . particularly high sodium contents". Write "some carry percent-level sodium salts, mainly NaCl".

4. **4. Sintering of plume deposits, and why conductivity** (tex L263, new on 10-15)

   Where on Enceladus deposits sinter, how fast against burial, and why their conductivity matters.

   - [written] Drafted along the plan: Choukroun's experiment, the Tiger Stripes, burial, Byrne et al.'s conductivity range and why it matters.
   - [correct] [lit. check F6] Numbers and wording, all checked against the PDFs:
     - "may reach up to 180 K": Spencer & Nimmo (2013, p.8): "surface temperatures within the troughs reach at least 180 K over regions tens of meters wide".
     - "little to no sintering is expected": Choukroun et al. (p.7) predict "little to no strengthening"; they measure strength.
     - "0.01—1.0 mm yr^{-1}" (Kempf; Southworth): the range is Choukroun et al.'s summary (p.7: "on order of 0.01 to 1 mm/yr (Kempf et al., 2010; Southworth et al., 2019)"); Kempf et al. give "0.5 mm per year in the immediate vicinity of the jet sources" (p.10). Cite Choukroun with them. The rates assume compact ice (Southworth et al., p.3: "a lower bound on deposition").
     - "so fresh deposits bury grains as they sinter" needs Choukroun et al. (p.7: "competition between strengthening of existing plume deposits and covering by new fresh and loose particles").
     - "Sintering that strengthens a deposit will also cause changes to . . . thermal conductivity" needs a citation: Molaro et al. (p.23): neck growth causes "an increase in the thermal conductivity of the aggregate".
     - Three `\citet` that should be `\citep`; "preformed", "grows linear", "with about 15 years", "expected be greatest"; en dash in the range.
   - [correct] [lit. check F11] `byrneRadarAttenuationSubsurface2024` is the preprint; use the JGR Planets article, doi:10.1029/2024JE008615. Their conductivities are for an airless surface; ours are with air at 1 atm, so do not place our values on their scale.

Suggested text: Under typical Enceladus surface conditions, however, little to no strengthening is expected, and the deposits remain poorly consolidated granular packings (`choukroun_strength_2020`). Near the Tiger Stripes, surface temperatures reach at least 180 K (`spencerEnceladusActiveIce2013`), and there the deposits could consolidate within about 15 years (`choukroun_strength_2020`). These are also the regions of heaviest plume deposition, on the order of 0.01–1 $\mathrm{mm\,yr^{-1}}$ (`kempfHowEnceladusDust2010, southworthSurfaceDepositionEnceladus2019, choukroun_strength_2020`), so fresh grains bury a deposit while it sinters (`choukroun_strength_2020`). As the necks grow, the conductivity of the deposit rises (`molaroMicrostructuralEvolutionWater2019`).

5. **5. Sintering as part of metamorphism** (tex L266; Figure 1 at L271)

   Neck growth between grains, which transport route dominates, earlier work.

   - [written] Choukroun is out of the vapor-dominance sentence; the snow-metamorphism examples are gone; the exponent sentence is in, with Hobbs & Mason, Kingery, Thomas and Demmenie in the `.bib`; Molaro et al.'s grain-pair measurements are named; Figure 1's caption is vapor-only.
   - [correct] [lit. check F4] **The exponent sentence cites the wrong sources.** Hobbs & Mason's model, with vapor diffusing through air to the neck, gives 1/5, not 1/3 (Blackford 2007, p.16: "Hobbs and Mason's modified model gave values of $n = 5$ and $m = 3$ for vapour transport"; their PDF is not in the library). The 1/3, 1/5, 1/7 set is Kingery (1960, Table I) and Demmenie et al. (2025, Table 1). Thomas et al. give "$n = 3$ for vapour transport, $4 \leq n \leq 5$ for surface diffusion" (p.3); cite them for what they measured: "a neck growth proportional to $t^{1/3}$, i.e., the sublimation and recondensation process predominate at $-20\,^{\circ}\mathrm{C}$" (p.4).
   - [correct] [lit. check F4] "n identifies the dominating mechanism": Maeno & Ebinuma (1983, p.4) call that "misleading because the sintering rate is usually the sum of contributions from several mechanisms". And name the debate: Kingery measured "$(x/r) \sim t^{1/6.9}$, in close agreement with … surface diffusion" (p.5); Demmenie et al. (p.2) attribute it to "an underestimation of the relative humidity".
   - [correct] [lit. check F4] "Other modes … operate over planetary timescales (Molaro)" reverses Molaro et al. (p.30): at 200 K neglecting surface diffusion changes their τ_{50} by "∼3,700%", and they "recommend always including the surface diffusion term … in planetary applications". At 250 K it matters only for the smallest grains (13% for 1 μm; "no significant effects" for 100 μm). Drop "or operate over planetary timescales"; this also supports not extrapolating to Enceladus temperatures.
   - [correct] The vapor-dominance sentence has no citation: Thomas et al. (1994, p.4) and Demmenie et al. (2025) are the direct ice measurements.
   - [add] **What Molaro et al. measured** is still thin: neck growth between grain pairs in a sealed cryostage at −20 and $-5\,^{\circ}\mathrm{C}$. They report no exponent; 0.23 ± 0.03 is our fit to their points.
   - [add] **Describe Demmenie et al.'s experiment in one sentence.** In ¶5 they are only one key in a list of four citations, but Section 3.2 tests the model against their numbers, so the reader needs to know what they measured: two ice spheres brought into contact in a box held at ice saturation, at $-3\,^{\circ}\mathrm{C}$, with exponents 0.26–0.33 (p.3), and "∼1/7" when "we deliberately set the humidity too low and let the ice sublimate" (p.3).
   - [add] **Scope: rounded grains, not fresh snow** (discussed 10-16). Molaro et al. (spheres sprayed into liquid nitrogen, p.15), Demmenie et al. (ice spheres) and our aggregates (discs

in point contact) all start from rounded grains. None of them is snow, fresh or aged, so they cannot be said to use "snow that has already metamorphosed". What can be said: fresh snow first rounds, and that stage is fast; Kaempfer & Schneebeli (2007) saw "the transition from the dendritic form of the fresh snow at time zero to rounded grains a month later" (p.5), the first phase of "a two-stage process" (p.1). Sphere experiments and our model describe the second stage, sintering of rounded grains. Our results do not cover the first stage, so they cannot claim it. Put this in Section 2.1 (Scope) and Section 5.6.

- **[wording]** "an adapted model sintering model originally used in metallurgy first proposed by": Molaro et al. made "No modifications" to it (p.5), and Swinkels & Ashby's is a "second report"; "applied the sintering model of `swinkelsSecondReportSintering1981`, developed for metals". "Sintering has been studied extensively ..." has no citation: Blackford (2007) is a review of ice sintering. The last sentence (Choukroun) repeats ¶4: cut.
- **[wording]** "In a laboratory setting moves tends be redistributed"; "Neck growth tend s"; w is "the neck width" here but "a neck of radius $w(t)$" in the Figure 1 caption: width throughout. Caption: "fo".

  Suggested text: In laboratory experiments, mass reaches the neck mainly by vapor transport: ice sublimates away from the neck and desublimates on it (`thomas_crushing_1994`, `demmenieIceSinteringSublimation2025`), as shown in `fig:Figure1_mechanisms_aggregate`. At these temperatures, other routes matter only for the smallest grains, but surface diffusion gains weight as temperature falls (`molaroMicrostructuralEvolutionWater2019`). Neck growth often follows $w \propto t^n$. When attachment at the ice surface sets the rate, vapor transport gives $n = 1/3$, while volume and surface diffusion give $1/5$ and $1/7$ (`kingery_regelation_1960`, `demmenieIceSinteringSublimation2025`); when vapor must diffuse through air to reach the neck, vapor transport also gives $n = 1/5$ (`blackfordSinteringMicrostructureIce2007`). Several mechanisms act at once, so the exponent alone does not identify the dominant one (`maenoPressureSinteringIce1983`). `kingery_regelation_1960` measured $n \approx 1/7$ and attributed it to surface diffusion, whereas `demmenieIceSinteringSublimation2025` measured n between 0.26 and 0.33 at saturation and close to $1/7$ when the humidity was too low and the ice sublimated, and attributed Kingery's value to an underestimated humidity.

Figure 1 **[written]** placed. Caption: see "Figure captions" after Section 6.

6. **6. Attachment kinetics** (tex L276)

Supersaturation, the Hertz–Knudsen law, and the condensation coefficient.

- **[written]** Libbrecht & Rickerby (2013) now cited for the measurements, on basal and prism facets; the layer-nucleation wording; the form of α_c; the range cited; a closing sentence on the constant α_c; "The growth rate of ice from vapor depends on".
- **[correct]** **[lit. check F2]** **"$\alpha_c = 10^{-3}$ to force a diffusion limited growth regime" is backwards.** A smaller α_c slows attachment and moves growth toward attachment control. Bouvet et al. (2024, p.5) place the transition at "10^{-3} for small grains (∼0.1 mm)", about the diameter of our grains ($\bar{R} = 50$ μm). The draft's Section 5.1 (tex L502) already says "attachment limited regime"; Section 3.2 (tex L455) says "vapor-limited" and needs the same fix.
- **[correct]** **[lit. check F8]** Libbrecht & Rickerby do not state the range 10^{-3}–10^{-1}; Braun et al. (2024, p.12) call it "the commonly found range", "also used by Kaempfer and Plapp (2009)". Cite those two; "bounded" → "commonly used". $\alpha_c = \exp(-\sigma_0/\sigma)$ is not an Arrhenius function (its argument is $1/\sigma$, not $1/T$); only σ_0 depends on temperature; "surface saturation" → "supersaturation".

- [wording] "led an experiment in near vacuum": "measured at reduced pressure". "and thus slowing attachment": "which slows attachment".

  Suggested text: Values of α_c between 10^{-3} and 10^{-1} are commonly used (`kaempferPhasefieldModelingDry2009, braunRigorousApproachSpecific2024`). In the layer-nucleation model of `libbrechtMeasurementsSurfaceAttachment2013`, $\alpha_c = A\exp(-\sigma_0/\sigma_{\text{surf}})$, where the critical supersaturation σ_0 depends on temperature and differs between the basal and prism facets. In this study we hold α_c constant. We use $\alpha_c = 10^{-1}$, the upper end of the range, to compare with the experiments of `molaroMicrostructuralEvolutionWater2019`, and $\alpha_c = 10^{-3}$ for the saturated grain pair and the aggregates. For grains of about 0.1 mm, $\alpha_c = 10^{-3}$ lies near the transition between attachment-limited and diffusion-limited growth (`bouvetMultiscaleModelingHeat2024`).

7. **7. Metamorphism beyond a pair; earlier models** (tex L279)

   From a grain pair to an aggregate, and the models that exist.

   - [written] Braun et al. described and the α_c fits stated as suggested; Libbrecht (2017) out of the DSM list; "all stages of DSM"; "to better predict".
   - [correct] [lit. check F10] "Many experiments attempt to parameterize" cites Colbeck (1983), a theory paper, and Libbrecht (2017), a review: cite Libbrecht & Rickerby (2013).
   - [add] (optional) End on Braun et al.'s reading, which is the evidence for our own use of α_c: they call it an "effective condensation coefficient" (p.13).
   - [wording] "x-ray compute microtomography": "X-ray computed microtomography". "were the first to attempt to use": "were the first to use". "steady state" (adjective): "steady-state".

8. **8. What is not known** (tex L282)

   The gap the paper addresses.

   - [written] "affects"; "bulk properties of porous ice"; the paragraph now ends on the paper's own gap; Calonne et al. (2011) and a gradient sentence added.
   - [correct] [lit. check F1] **The gap is partly closed already.** "How isothermal metamorphism influences the macroscopic properties in the absence of compaction is still unquantified" is contradicted by Bouvet et al. (2022), cited in this paragraph: they computed k_{eff} on snow evolved isothermally by their phase-field model (p.12: "conductivity increases from 0.078 to 0.081 $\mathrm{W\,m^{-1}\,K^{-1}}$"; "changes are rather small"), and could not separate microstructure from density (p.15: "the impact of the microstructural changes is not visible"). What remains is ours: grains sintering from point contacts at fixed porosity, conductivity against SSA, and the effect of temperature and porosity.
   - [correct] [lit. check F1] The gradient sentence follows Calonne et al. (2011), a survey of 30 samples with no time series. Its evidence is Calonne et al. (2014): SSA fell "from 27.7 to 13.4 $\mathrm{m^2\,kg^{-1}}$" (p.9); conductivity showed "a decrease between 0 and 73 h, followed by an increase" (p.16), "an overall increase ... at a constant density" in their words. Schneebeli & Sokratov (2004, p.10) saw a rise "by about 60%". Our reconstruction from their Table 1 (below) has the orientation average ending near its start while k_z rose; say that the loss of SSA did not by itself raise conductivity, not that it never did.
   - [correct] [lit. check F10] "centered around the snow community here on earth" cites Flin et al. (2003, 2004), who report microstructure, not bulk properties: cite Calonne et al. (2011, 2014). "here on earth": "on Earth".

- [add] **Why SSA**, one sentence: SSA is measured in the field and laboratory (Matzl & Schneebeli 2006; Gallet et al. 2009; Proksch et al. 2015), so a law in SSA can be applied. Still missing.
- [add] **What the conductivity here includes**, one clause: under a gradient, vapor transport adds to the apparent conductivity (Fourteau et al. 2021); ours is conductive only, which is exact for isothermal sintering. Still missing (here or in Section 2.3).
- [from the notes] **Background on snow conductivity** (for the two sentences above; from the earlier outline). Calonne et al. (2011): full conductivity tensor from tomography of 30 samples, a regression on density ($r = 0.985$), scatter from anisotropy (k_z/k_{xy} 0.7–1.5). Calonne et al. (2014): rounded grains, 294 $\mathrm{kg\,m^{-3}}$, 43 $\mathrm{K\,m^{-1}}$ at $-4\,^{\circ}\mathrm{C}$ for 3 weeks; density constant; SSA 27.7 → 13.4; density-corrected k_z 0.18 → 0.17 → 0.23, k_{xy} 0.20 → 0.15 → 0.18 (our reconstruction, ±0.01); k_z/k_{xy} 0.90 → 1.34; they predict k from porosity and correlation lengths, never from SSA. Calonne et al. (2019): k_z/k_{xy} tracks the correlation-length ratio; at low density "the few bonds between ice grains … act as heat paths" (p.5).

  Suggested text: Under a temperature gradient, `calonneStudyTemperatureGradient2014` followed a snow layer at constant density for three weeks: its SSA halved, while its conductivity first decreased, as bonds between grains sublimated, and then increased, mostly in the vertical direction, a two-stage change that `schneebeliTomographyTemperatureGradient2004` had also seen. Losing surface area did not by itself raise the conductivity. Under isothermal conditions, `bouvetSnowEquiTemperatureMetamorphism2022` computed the conductivity of snow evolved by a phase-field model and found only small changes, which followed small variations in density. Neither study follows a packing of ice grains as it sinters from point contacts at fixed porosity, relates the conductivity to the SSA, or asks how temperature and porosity change that relation.

9. **9. What follows** (tex L285)

   The order of the paper and its findings.

   - [written] The two findings are rewritten: the conductivity–SSA relation is independent of temperature, and for $\varphi \leq 0.375$ it is a power law with a shared exponent.
   - [correct] Porosity: say also that the level falls with porosity (about threefold from $\varphi = 0.275$ to 0.475); "same exponent" alone hides it.
   - [correct] "an updated phase-field model introduced by Moure & Fu": Moure & Fu (2024) is a wet-snow model; "the ice–air form of the phase-field model of Moure & Fu (2024)".
   - [correct] "use their results to compare with their model": "compare our model with their measurements".
   - [correct] "a diffuse interface adapted homogenization technique initially outlined by Auriault (1991)": "a homogenization method (`auriaultHeterogeneousMediumEquivalent1991`) adapted to the diffuse interface".
   - [add] **Announce the two-grain results** in one sentence: vapor transport alone gives 41% of the neck growth Molaro et al. measured at $-20\,^{\circ}\mathrm{C}$, and the exponent Demmenie et al. measured at saturation (0.29 against 0.26–0.33).
   - [add] **The temperature result is not announced**: temperature from -40 to $-5\,^{\circ}\mathrm{C}$ only rescales time (Section 5.2). One clause.
   - [add] **Scope, in one clause**: isothermal, air at 1 atm, two dimensions, temperatures a laboratory can reach.
   - [add] **What is new**, one clause: the thin-interface kinetics and the homogenization in the diffuse interface (tensor law; prior art Nicoli et al. 2011).

- [wording] "In what follows" opens this paragraph and the last sentence of ¶8: once. "we will present": "we present". "over 2D packings ... over 30-day time periods": "in two-dimensional packings of ice grains for 30 days". "tracks the temperature and water vapor density fields": and the phase field.

2 Modeling Approach

Draft heading: 2 Modelling Approach.

1. **Section opening** (tex L293)

   Says the model follows Kaempfer & Plapp and Moure & Fu.

   - [written] The novelty sentence ("The contribution here is the application of the phase-field model, not the numerical methods"); the homogenization lineage (Auriault 1991; Calonne et al. 2011, 2014, 2019).
   - [add] [lit. check F10] Calonne et al. (2014, p.6) cite the Auriault et al. (2009) book for this boundary-value problem; add it next to Auriault (1991). The book is in the library; it is not in the `.bib`.
   - [add] One sentence of lineage from the Discussion's second paragraph can move here.

2.1 Phase-Field Model of Dry Snow Metamorphism

2. **What a phase field is** (tex L297)

   General description of the method.

   - [wording] "representing the volume fraction of phase l at a given point ... and evolves through time": the sentence changes subject; "which evolves in time".
   - [trim] Textbook material; one sentence.

3. **Unknowns** (tex L300)

   Lists the four fields.

   - [written] ϕ_a is "air".
   - [correct] **Temperature in °C (user, 10-16).** T is in °C, which is how every result is reported; the solver works the same way (`RhoVS_I: temK = tem + 273.15`). Introduce the absolute temperature $T + T_0$ ($T_0 = 273.15$ K) once, here, and use it only where a formula needs kelvin: Eq. (3) (Wexler), Eq. (5) (D_v), Eq. (7) (β_{sub}, $k_B(T + T_0)$) and d_0. The conservation equations (1b), (1c) and (11) involve only ∂T and ∇T, so they are unchanged. Table 1 [written] 10-16.

     Suggested text: The temperature T is expressed in degrees Celsius; where a relation requires the absolute temperature we write $T + T_0$, with $T_0 = 273.15$ K.

   - [wording] "derived from the free energy potential and allows": "that allows".
   - [trim] The last two sentences ("defined throughout the domain", "benefit from the phase-field formulation") can go.

4. **Scope** (to write)

   Says which part of dry snow metamorphism the model contains.

- [add] The model carries sintering by vapor transport: sublimation, diffusion through the pore, desublimation.
- [add] Not in it: surface diffusion, grain-boundary processes, densification, overburden. Define desublimation.
- [add] Vapor-transport sintering does not densify: grain centres do not approach, so porosity is constant in every run.
- [add] Hence the laboratory analogues are unloaded ice-sphere samples, not natural snow.

5. **Model equations and the phase equation** (tex L304–311)

   Eq. (1) and the meaning of its terms.

   - [written] Eq. (1b) carries ρ_i; "ε controls the width"; "Allen-Cahn".
   - [correct] [lit. check F9] M_{sub} and α_{sub} now point to the appendix "Model Scaling Parameters" (tex L527), which is an empty heading; label typo `Sclaing`. Content: $M_{\mathrm{sub}} = \varepsilon/(3\tau_{\mathrm{sub}})$, α_{sub}, β_{sub}, $\Delta\beta$, τ_{sub} (now in the notes to Table 1).
   - [correct] Eq. (1c) as printed has no ξ_v, but Tables 3 and 4 list $\xi_v = 10^{-3}$. Show it or explain it in one sentence.
   - [wording] "The product … localize this phenomena": "localizes this term".

6. **Saturation vapor density** (tex L313–319)

   Eqs. (2) and (3).

   - [written] Eq. (3) is now Wexler's (1977) Eq. 54, as used by Kaempfer & Plapp (2009); "over ice"; "are".
   - [correct] [lit. check F9] Eq. (2) still writes P_a; the text says P_{atm}. "where $T_0 = 273.15$ K" follows Eq. (3), which does not contain T_0 yet. With T in °C, Eq. (3) reads $P_{vs}^I = \exp\left(\sum_{j=0}^{4} k_j (T+T_0)^{j-1} + k_5 \ln(T+T_0)\right)$, and the sentence then defines T_0 (snippet below).
   - [correct] [lit. check F9] **Table [written] 10-16:** Eq. (3) is Wexler (1977), Eq. 54 (p.11), coefficients K_j in his Table 2 (p.9); the solver's k_j are these rounded to four figures (+0.04 to +0.05% in P_{vs}; Wexler's Table 3 gives 103.276 Pa at −20 °C). Table 1's note b now cites Wexler Eq. 54 and lists the six values; "ASHRAE" is gone (ASHRAE uses Hyland & Wexler 1983, a different fit). **Open (text):** the sentence after Eq. (3).

     Suggested text: where $T_0 = 273.15$ K and P_{vs}^I in pascals (`wexlerVaporPressureFormulation1977`); the coefficients k_j are listed in `tab:material_properties`.

7. **Energy equation** (tex L321)

   Eq. (1b) and the mixture rules.

   - [correct] "latent heat released and absorbed during sublimation and desublimation, respectively": reversed. Sublimation absorbs heat; desublimation releases it.
   - [wording] "phase dependent": hyphenate.

8. **Vapor equation** (tex L331–335)

   Eq. (1c) and the diffusivity.

- **[correct]** With T in °C, Eq. (5) becomes $D_v = D_{v0}\left((T+T_0)/T_0\right)^{1.81}$, with D_{v0} the diffusivity at 0 °C. Tables 2–4 used T_0 for the run temperature in °C; since 10-16 they write T, so T_0 is only 273.15 K.
- **[wording]** "The rest of the thermophysical properties are shown in": "Material properties are listed in".
- **[add]** Massman (1998) for $D_v(T)$, as Moure & Fu do; not in the library, so not checked here.

Table 1 **[written]** Material properties, cited at tex L335.

2.2 Interface Kinetics and the Sintering Time Scale

Draft heading: 2.2 Gibbs-Thomson Condition.

1. **The interface condition** (L216–226)

   Eqs. (6) and (7).

   - **[wording]** "is commonly interchanged with the condensation coefficient": "is commonly expressed through". "molecular mass a water molecule" (mass of a water molecule).
   - **[add]** $d_0 = \gamma m/(\rho_{\text{ice}} k_B T)$ is defined only in Table 1.

2. **The thin-interface limit** (to write)

   What "under the conditions outlined by Kaempfer & Plapp" means.

   - **[add]** The model recovers Eq. (6) with a kinetic coefficient $\beta_{\text{sub}} + \Delta\beta$, $\Delta\beta = a_1 a_2\, \varepsilon\, (D_T^{-1} + D_v^{-1})\, \rho_{\text{ice}}/\rho_{vs}^I$.
   - **[add]** Its size (2% aggregates, 0.2% saturated pair, 23% Molaro pairs) goes in the Discussion, not here (user, 10-16): Section 5.1, "The unsaturated pair".
   - **[add]** **Note (user, 10-16).** The text will say the model reproduces Eq. (6) "under certain parameter regimes", as Moure & Fu (2024, p.5) do, and may list the conditions in an appendix. They are Kaempfer & Plapp's (2009, pp.8–9): $W \ll D_v\beta'$ and $W \ll (\kappa/C)\beta'$ (Eq. 43), $W \ll d_0/(\beta_0 v_n)$ (Eq. 45), and $W \ll 1/\mathcal{K}$. These are the four bounds `preprocess/comp_eps.py` applies to ε ([B-VAPOR], [B-HEAT], [B-KINETIC], [B-CURV]). Literature check: they are not isothermal conditions. Kaempfer & Plapp obtain the $\Delta\beta$ scaling by dimensional argument rather than a derivation for one-sided vapor diffusion (p.8), so the appendix should present $\Delta\beta$ as an estimate.

   Cite: Kaempfer & Plapp 2009 `kaempferPhasefieldModelingDry2009`; Moure & Fu 2024 `mourePhaseFieldModelWet2024`.

3. **The time scale** (to write here; now at L469–477 in the Discussion)

   Defines τ_{sub} and the sintering age θ.

   - **[add]** Move Eq. (τ_{sub}) from Section 5.1 to here, and define $\theta = t/\tau_{\text{sub}}$ here, not in Section 4.2.
   - **[add]** Two time scales of the same form $\ell^2\beta/d_0$: τ_{sub} ($\ell = \varepsilon$), the relaxation time of the diffuse interface; and $\tau_{\bar{R}} = \bar{R}^2\beta/d_0$, the time for a grain to change shape. $\tau_{\text{sub}} = \tau_{\bar{R}}/2500$ in every aggregate run.
   - **[add]** The 30-day runs end at $\theta = 43$ (−40 °C) to 1256 (−5 °C). Cite Table 2 here.

- [add] **Example paragraph (10-16).** One paragraph after Eq. (7) can replace three entries: "The thin-interface limit", "The time scale" and "The values used". The size of $\Delta\beta$ stays out of it (it goes in Section 5.1).

  Suggested text: Our model reproduces `eq:GibbsThomson` under certain parameter regimes (`kaempferPhasefieldModelingDry2009, mourePhaseFieldModelWet2024`), which require ε to be small compared with the radius of curvature of the ice and with the diffusion and kinetic length scales of the problem (`sec:Appendix_SclaingParameters`). For a finite ε, the interface moves with the kinetic coefficient $\beta_{\text{sub}} + \Delta\beta$, where $\Delta\beta$ is the thin-interface correction (`tab:material_properties`). This sets the characteristic time of sublimation in our model,

  $$\tau_{\text{sub}} = \frac{\varepsilon^2}{d_0}\left(\beta_{\text{sub}} + \Delta\beta\right),$$

  the time over which the diffuse interface relaxes. A grain of mean radius $\bar{R}$ changes shape over the time $\bar{R}^2(\beta_{\text{sub}} + \Delta\beta)/d_0$, which is $(\bar{R}/\varepsilon)^2 = 2500$ times τ_{sub} in every aggregate simulation. We therefore report time as $\theta = t/\tau_{\text{sub}}$: our 30-day simulations end at $\theta = 43$ at $-40\,^{\circ}\text{C}$ and at $\theta = 1256$ at $-5\,^{\circ}\text{C}$ (`tab:keff_temperature`). Since $\beta_{\text{sub}} \propto 1/\alpha_c$ (`eq:Beta_sub`), the condensation coefficient sets this time scale. We hold α_c constant with temperature: 1.020×10^{-3} in the aggregate simulations, 1.002×10^{-3} in the saturated grain pair, and 0.1 in the grain pairs of `molaroMicrostructuralEvolutionWater2019`, the upper end of the range reported in the literature (`libbrechtPhysicalDynamicsIce2017, bouvetSnowEquiTemperatureMetamorphism2022, braunRigorousApproachSpecific2024`). A temperature-dependent α_c would change the kinetics and the temperature together, and the effect of temperature alone could no longer be read from our simulations.

- [add] Check before pasting: the label of Eq. (7) (written here as `eq:Beta_sub`); the appendix label keeps the draft's spelling `Sclaing` until you fix it; Eq. (τ_{sub}) then leaves Section 5.1 (tex L507), and that paragraph refers back to it.

4. **The values used** (to write)

   The condensation coefficients and why they are constant.

   - [add] $\alpha_c = 1.020 \times 10^{-3}$ (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair), constant in temperature.
   - [add] Why constant: a temperature-dependent α_c would make a temperature sweep uninterpretable.

   Cite: range of α_c: Braun et al. 2024 `braunRigorousApproachSpecific2024`; Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Libbrecht 2017 `libbrechtPhysicalDynamicsIce2017`.

Table 2 [written] Temperature-dependent quantities, placed. [add] Not cited in the text.

2.3 Effective Thermal Conductivity by Homogenization

Draft heading: 2.3 Homogenizing for Effective Thermal Conductivity.

1. **Two scales and the microscale problem** (L228–238)

   Scale separation, periodicity, Eq. (8).

   - [correct] "without less of generality": periodicity is an assumption, not a convenience (and the phrase is "loss").

- [wording] "A smaller length scale l describing ..., and L describing ..." is not a sentence. "representative element volume" (elementary). "diffusive heat transfer" (conductive). "upscaling effective thermal conductivity" (computing the).

2. **Macroscale law and cell problem** (L239–247)

   Eqs. (9)–(11).

   - [correct] [lit. check F9] Eq. (11d) (tex L379) has $\mathbf{t}$ where it needs $\nabla\mathbf{t}$, as in Eqs. (11a, b) and in Calonne et al. (2014, p.6, their Eq. 15).
   - [wording] "are commonly referred to as co the solutions": the sentence is broken. "the corresponding macroscopic description is described as".
   - [trim] "Equations ... are then coupled with the boundary conditions ..." says the equations go with their own boundary conditions; cut.

3. **Single-domain form** (L248–260)

   Weak form and the indicator function, Eqs. (12)–(15).

   - [correct] The test function is scalar v in the text and bold $\mathbf{v}$ in Eq. (12). $K_\star$ in Eqs. (13)–(14) becomes K_* in Eq. (15).
   - [wording] "as diffuse interface model" (as a); "in 2.1" (Section 2.1); "We can the define" (then); "$\zeta = 1.0$" (1).
   - [trim] Eqs. (12)–(14) can be one sentence: summing the weak forms over the two subdomains, the interface terms cancel by flux continuity, and Eq. (15) follows.

4. **Diffuse form and the tensor law** (L261–275)

   Eqs. (16), (17) and why the conductivity is a tensor in the interface.

   - [correct] "interpolates between 0 and 1": between K_{air} and K_{ice}.
   - [wording] "along the our phase-field"; "$K_\parallel(\phi)$ and are $K_\perp(\phi)$ the"; "We then may write diffuse BVP" (the diffuse); "tonsorial" (tensorial); "interface width ε". ϕ and ϕ_i are used for the same field; $t_\star$ is not bold.
   - [add] Credit and what is new: Nicoli, Plapp & Henry (2011) define the law, show the cancellation analytically for a planar interface, and test it on a disk (contrast 2) and two half-disks (contrast 10); here it is applied to the cell problem at a contrast of 115 and verified on laminate and disk-array cases (supplement).
   - [correct] [lit. check F10] "both of order ε for a scalar conductivity law" (tex L421): a scalar law can cancel one of the two interface effects but not both; Nicoli et al. (p.1): "both effects can be eliminated simultaneously only if the mobility is allowed to become a tensor". Write "no scalar law removes both".
   - [add] Why it matters here: with a scalar law the bias reaches tens of percent and grows with ε SSA, so it would drift as the aggregate sinters.
   - [add] The cell problem is solved on frozen snapshots of ϕ.
   - [add] **Example paragraphs (10-16): credit and what is new.** Checked against Nicoli et al. (2011). Theirs: the two effects and the tensor (abstract, p.1); the law is their Eq. 17, with $M_\parallel$ the direct (Eq. 9) and $1/M_\perp$ the inverse (Eq. 16) interpolation; the cancellation shown for a planar interface (pp.2–3); numerical tests of one disk ($M_2/M_1 = 1/2$) and two half-disks ($M_2/M_1 = 0.1$) in a square with Dirichlet and no-flux boundaries, a fixed ϕ, and convergence "almost quadratic in ε" (p.4). They do not treat homogenization or an effective

conductivity. Ours: the law in the periodic cell problem, a contrast of 115, ϕ taken from the metamorphism model, and the verification. Numbers below: `studies/keff_sharp_limit/` (RECORD.md, 2026-09-22/23).

Suggested text: The phase-field problem is the sharp problem `eq:weak_sharp_interface` with $K_\star$ and $\mathbf{t}_\star$ replaced by a conductivity $\mathbf{K}(\phi, \nabla\phi)$ and a corrector $\mathbf{t}(\phi)$ defined throughout the cell. A diffuse interface of finite width does not behave as a sharp one: it adds an excess conductance along the interface and an excess resistance across it, both of order ε (`nicoli_tensorial_2011`). A scalar conductivity can remove one of these, but not both; an arithmetic mean removes the excess conductance and a harmonic mean removes the excess resistance. `nicoli_tensorial_2011` showed that both are removed if the conductivity is allowed to become a tensor in the interface, arithmetic along it and harmonic across it, and we adopt their law: [Eqs. (16a, b) as in the draft]. In the bulk, $K_\parallel = K_\perp = K_\star$ and $\mathbf{K}$ reduces to $K_\star\mathbf{I}$. The diffuse form of the cell problem is then [Eq. (17) as in the draft], where the interface conditions are satisfied by construction.

Suggested text: `nicoli_tensorial_2011` derived this law for transport across a planar interface and tested it on single inclusions at conductivity ratios of 2 and 10. We apply it to the periodic cell problem of homogenization, at the ratio $K_{\text{ice}}/K_{\text{air}} = 115$, on microstructures produced by the metamorphism model itself, so that $\mathbf{K}_{\text{eff}}$ is computed on the same phase field without segmenting it. At this contrast the choice matters. For a periodic array of ice disks the arithmetic law overestimates $\mathbf{K}_{\text{eff}}$ by 1.9% at an ice fraction of 0.05 and by 39% at 0.50, while the tensor law stays within 1% (Supporting Information). Because the error of a scalar law scales with ε times the specific surface area, it would also change as the aggregate sinters, and so bias the trends we report.

- **[add]** Check before pasting: the disk-array numbers are for single cylinders, 6 radii, 3 values of ε/R (RECORD.md L41); the Supporting Information section does not exist yet. Ettrich et al. (2014) applied the law to heat conduction; cite only if it is added to the `.bib` (not in the library).

Cite: Nicoli, Plapp & Henry 2011 `nicoli_tensorial_2011`; Auriault 1991 `auriaultHeterogeneousMediumEquivalent1991`; Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

5. **What is reported** (to write)

   Defines the quantities the results plot.

   - **[add]** $\mathbf{K}_{\text{eff}}$ in the diffuse form, the one evaluated (Eq. 10 is the sharp form only).
   - **[add]** The scalar $k_{\text{eff}} = (k_{xx} + k_{yy})/2$, now defined only in the Figure 6 caption.
   - **[add]** SSA: how it is measured from ϕ, and its units. It is used throughout Section 4 and defined nowhere.

   Cite: for computing SSA from a field, possibly Flin et al. 2014 `flinComputationsSpecificSurface2014` (check that the method matches).

Figure 2 **[written]** placed. **[add]** Not cited in the text. Caption: see "Figure captions" after Section 6.

2.4 Numerical Implementation

1. **Discretization and time stepping** (to write; heading only at L276)

   Enough for someone to reproduce the runs.

   - **[add]** Quadratic C^1 splines; element size $\varepsilon/\sqrt{2}$.

- [add] Adaptive step capped at $2\,\tau_{\mathrm{sub}}$ in the aggregates. The grain-pair runs use larger caps ($5\ \tau_{\mathrm{sub}}$ for the saturated pair; 30 s, i.e. 22 τ_{sub}, for Molaro $-20\,^{\circ}\mathrm{C}$); say so.
- [add] Conductivity sampled every step for $\theta < 11$, then at every 0.1% drop in SSA.

2. **Checks** (to write)

   The results do not depend on domain size, time step or sampling.

   - [add] One sentence each, pointing to the supplement: domain size (Fig. S1); interface parameter (Fig. S2), including that the first day depends on ε and later rates do not; time step and sampling (Fig. S3).

Table 3 [written] Complete parameters, placed. [add] Not cited in the text. *Open: keep it here, or the short version here and this one in the supplement.*

2.5 Simulation Design: Grain Pairs and Aggregates

1. **Grain pairs** (to write; heading only at L277; material now at L301–313 and L326–331)

   What the three two-grain simulations are.

   - [add] Axisymmetric about the line of centers; fixed temperature and humidity on the outer boundary; how the neck width w is measured.
   - [add] The two Molaro geometries and the equal-grain pair ($R = 84.4$ μm, $-20\,^{\circ}\mathrm{C}$, 100 h).
   - [add] Every pair starts from two overlapping spheres with a neck 28 μm wide, not from grains that just touch. Say so here; it matters for the exponents (Section 3.2).

Table 4 [written] Two-grain parameters. [add] Add a column for the saturated pair. [correct] Simulated time at $-5\,^{\circ}\mathrm{C}$ is 68 min, not 2 h.

2. **Aggregates** (to write; material now at L343–352)

   What the 125 aggregate simulations are and how the packings were made.

   - [add] 2 mm square, periodic in both directions; circular grains, log-normal radii with mean 50 μm, placed by drop-and-roll deposition.
   - [add] Five porosities, five independent packings each, the same packings at all five temperatures: 125 runs of 30 days; uniform temperature, vapor initially at saturation.
   - [add] **[lit. check F5]** Domain size $L/\bar{R} = 40$, [del: at the 1–2 mm representative volume of Calonne et al. (2011).] from our domain-size test (F5); Calonne et al. (2011) give 2.5–5.5 mm for natural snow. Packings are accepted only if contacts across the periodic seam are as dense as in the interior.

   Cite: representative volume: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

3. **Why periodic, why two dimensions** (to write)

   Justifies the two choices a reviewer will question first.

   - [add] Periodic: the domain stands for the interior of a deposit, no wall physics has to be invented, and the system is closed, so curvature alone drives transport.
   - [add] Two dimensions: cost (now at L290–291, in the wrong place); what it costs is discussed in Section 5.6.

3 Neck Growth Between Two Ice Grains

1. **Section opening** (L279–294)

   Announces the findings.

   - [correct] It was written as the opening of a single Results section and still announces the aggregate findings of Section 4.
   - [correct] "independent of the mean temperature as well as the initial porosity"; "our model's ability to capture sintering dynamics for varying temperatures": neither is what the results show.
   - [correct] "theorized trends from Demmenie et al.": they measured the exponent as well. "dominated by vapor transport": the $t^{1/3}$ law is for sublimation and desublimation limited by attachment.
   - [wording] "simulation results, which captures" (capture); "related the evolution" (to the); "microstructures SSA" (microstructure's).
   - [trim] Two sentences: before using the model on aggregates, we ask how much of measured neck growth it reproduces, with one experiment away from saturation (the rate) and one at saturation (the exponent).

3.1 Comparison With Unsaturated Experiments

1. **Why these experiments** (L295–300)

   Why Molaro et al. is the comparison.

   - [correct] "relatively well-controlled": the humidity was not reported and the grains sublimated; Section 5.1 argues the conditions were not controlled at the level of the driving force. Say "among the few that isolate a grain pair at these sizes and temperatures".
   - [wording] "are good candidate" (a good candidate); "validating" (testing); "that exist that isolates" (isolate).

2. **What is fitted and what is not** (L301–313)

   The set-up of the $-20\,°C$ run.

   - [correct] "The only free-parameter in our model is … β_{sub}": the humidity is fitted too.
   - [correct] "over a 78 minute period": Molaro's text describes their Figure 11 as 57 min apart; our digitized series runs to 78 min. Check which figure the series comes from and cite it.
   - [correct] "within a sealed cryostage": their text says the sample area was not sealed off from the chamber, and the grains shrank.
   - [wording] "fig. 11" (their Figure 11); "We assumed a value … and was at the upper end" (state $\alpha_c = 0.1$); "(cite other works as well)"; "table table 4".
   - [add] The inferred humidity, 0.99715 at $-20\,°C$ and 0.99674 at $-5\,°C$, from the shrinkage of the large grain.
   - [trim] Geometry and boundary conditions move to Section 2.5.

3. **Result** (L314–324)

   Figure 3: model against measurement.

- [wording] "Panel (a) we see" (shows); "the a digital replicate"; "corresponds final temporal data point"; "over approximates and under approximates" (overpredicts, underpredicts); "will discussed" (be); "section section 5.1"; "that they cryostage chamber".
- [add] The numbers: 41% of the measured neck growth at $-20\,^{\circ}\mathrm{C}$ over 78 min; 170% at $-5\,^{\circ}\mathrm{C}$ over 48 min (52.1 against 44.0 μm).
- [add] Panels (c), (d): the shrinkage is what the humidity was fitted to, so they are a calibration, not a test.
- [add] Molaro's conventions are used (neck width; time from the first measurement). Their own model matches through its vapor term, which assumes $\alpha_c = 1$ and no gas diffusion.

Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

Figure 3 [written] placed. Caption: see "Figure captions" after Section 6.

3.2 Growth at Saturation

1. **Set-up and result** (tex L454–456; **reframed 10-15** for the tangent-start run)

   Two grains brought into contact in saturated air: the model gives the measured exponent.

   - [add] **Framing.** Section 3.2 is now the positive test of the two-grain comparison. At saturation the grains neither grow nor shrink (radius +0.08% over 150 h), so the neck is the only thing that grows, and the model gives the exponent Demmenie et al. measured and the sharp-interface law predicts. Section 3.1 (Molaro et al.) remains the test of the *rate*; Section 3.2 is the test of the *exponent*.
   - [add] **Set-up, as run.** Two grains, $R = 84.4$ μm, just touching at $t = 0$; $-20\,^{\circ}\mathrm{C}$; $\alpha_c = 1.002 \times 10^{-3}$; vapor saturated over the grains' own curvature, $\rho_v = \rho_{vs}^{I}(1 + 2d_0/R)$; 3-D axisymmetric; 150 h.
   - [add] **Result.** Fitting $w = C(t + t_0)^a$ after the relaxation period (20 min, $11\,\tau_{\mathrm{sub}}$) gives $a = 0.289$ ($t_0 = 0.6$ h, rms 0.17%); the Kuczynski form gives $1/m = 0.290$; fixing $a = 1/3$ raises the rms to 2.6%. Starting the fit later: 0.289 (1 h), 0.285 (13 h), 0.281 (32 h), 0.279 (63 h), 0.277 (88 h). The neck reaches 61.2 μm ($w/2R = 0.36$).
   - [add] **What it is compared with.** Demmenie et al. measured 0.26–0.33 at saturation (p.3), on spheres at $-3\,^{\circ}\mathrm{C}$: the exponent is compared, not the rate. The sharp-interface law integrated for this geometry and α_c gives 0.29 (Appendix B); 1/3 is its limit for small necks, and the neck here is not small.
   - [add] **Why a small α_c gives $t^{1/3}$, and a large one $t^{1/5}$** (the reader will ask; derivation in Appendix B). Both are vapor transport; they differ in which step limits it. Vapor leaves the convex grain surface and arrives at the concave neck, driven by the curvature of the neck fillet, $\Delta\sigma \approx d_0/\rho$, with fillet radius $\rho \approx x^2/2R$. It passes two resistances in series: attachment at the ice surface, $\beta_{\mathrm{sub}} \propto 1/\alpha_c$, which does not depend on size, and diffusion through the air over a length of order ρ, which grows with the neck.
     - Attachment-limited (small α_c): $v_n \approx d_0/(\beta_{\mathrm{sub}}\rho) \propto R/x^2$, so $x^2\,dx \propto dt$, $x^3 \propto t$, $w \propto t^{1/3}$.
     - Diffusion-limited (large α_c): $v_n \propto d_0/(k_D\rho \cdot \rho) \propto R^2/x^4$, so $x^5 \propto t$, $w \propto t^{1/5}$.

     A resistance that grows as the neck grows slows the neck faster, hence the smaller exponent. The classical 1/3 (Kingery 1960, Table I: "Evaporation condensation, $(x/r)^3 = B(T)t/r^2$") assumes the gas imposes no resistance: Kuczynski's model "assumed the mean free path

... of water molecules at atmospheric pressure in air to be large compared with the neck radius" (Blackford 2007, p.16). In air it is not ($\sim$0.1 μm), so Hobbs & Mason put in diffusion through air and got $n = 5$ (same page). A small α_c makes attachment so slow that the gas resistance becomes negligible even in air, which restores the condition the 1/3 law assumes. Here gas diffusion is 0.4–2% of the resistance at $\alpha_c = 10^{-3}$; our neck ODE gives $a \approx 0.20$ at $\alpha_c = 0.1$ and 0.29–0.30 at 10^{-3} (`studies/molaro_2019/envelope_ode/`), and the run gives 0.289. It is below 1/3 because the neck is not small (x/R up to 0.36: the $2/R$ term and the exact fillet slow it).

- [add] **Grain scale against neck scale** (so this does not contradict Intro ¶6). Bouvet et al. (2024, p.5) place the transition near $\alpha_c = 10^{-3}$ for 0.1 mm grains, with the diffusion length set by the grain size. At the neck the diffusion length is the fillet radius, a few μm, so the neck stays attachment-limited up to larger α_c than whole-grain growth does. In the aggregates the exchange is between grains, and the two resistances are comparable ($\Lambda \approx 2.8$).
- [add] **Where it goes:** the manuscript text is two sentences in Discussion 5.1 ("The saturated pair, and exponents"). This bullet and Appendix B are the background for it, not text for the paper.
- [correct] [lit. check F2] "a vapor-limited regime ... governed by vapor transport" (tex L455): the reverse; Bouvet et al. (2024, p.5) put the attachment–diffusion transition near 10^{-3} for 0.1 mm grains. At $\alpha_c = 10^{-3}$ growth is limited by attachment (gas diffusion adds 0.4–2% to the resistance), which is the regime the $t^{1/3}$ law describes.
- [correct] Numbers in the draft that describe the earlier run: $t = 100$ h $\rightarrow$ 150 h; fit window [20 min, 100 h] $\rightarrow$ [20 min, 150 h]; $a = 0.21 \rightarrow 0.29$; "briefly brought into contact" $\rightarrow$ "just touching".
- [correct] "The red dashed curve is the theoretical fit": it is the best fit with the exponent fixed at 1/3.
- [wording] "in attempts to compare" (to compare); "allowed for the to sinter"; "an condensation"; "1.002$e-03$" (1.002×10^{-3}); "Similarly to".

Suggested text: We performed one further simulation to compare the exponent of neck growth with that measured by `demmenieIceSinteringSublimation2025`. Two ice grains of radius $R = 84.4$ μm, initially just touching, sinter at $T = -20$°C for 150 hours in vapor saturated over the grains' own curvature, so that the grains neither grow nor shrink and only the neck evolves. We use $\alpha_c = 1.002 \times 10^{-3}$, for which growth is limited by attachment at the ice surface. After the initial relaxation period, the neck width follows $w = C(t + t_0)^a$ with $a = 0.29$ (`fig:Figure4_saturated_neck_growth`), and the exponent changes by less than 0.02 as the start of the fit is moved from 1 to 88 hours. This lies within the range 0.26–0.33 that `demmenieIceSinteringSublimation2025` measured at saturation, and matches the value 0.29 that the sharp-interface law gives for this geometry (Appendix B); the small-neck limit of that law, $a = 1/3$, fits our data less well.

Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`.

2. **The starting neck** (to write; one or two sentences here, the rest in Appendix B)

Why the run starts from touching grains.

- [add] An earlier run started from a pre-made neck (two overlapping spheres, a 28 μm neck) and gave $a = 0.21$–0.24 under the same conditions. The low exponent came from that initial condition: a fit must assume how long a pre-made neck had been growing, and the free fit

picks 4.8 h where the prediction says 16 h. Report it in Appendix B as a caution on initial conditions, not in the main text.

Open: the Molaro runs (Section 3.1) still start from a pre-made neck, because a tangent start traps an air pocket at the neck at that resolution (pilot of 10-15). Decide how Sections 3.1 and 5.1 qualify the 0.135 exponent at $-20\,^{\circ}C$, which may carry the same bias, and whether the 41% rate comparison, anchored on Molaro's first measurement, is affected.

Figure 4 [written] placed. Caption: see "Figure captions" after Section 6.

4 Thermal Conductivity of Sintering Aggregates

4.1 Microstructural Evolution

1. **Set-up and the gallery** (L343–361)

   States the simulation matrix, then describes Figure 5.

   - [correct] "mesh of size 2829×2928": 2829 both ways.
   - [correct] "We initially consider a single 2D microstructure": the previous sentence states 125 runs, and Figure 5 shows five.
   - [correct] "periodic boundary conditions for our equation (1b) and (1c)": all three fields are periodic.
   - [correct] "By the 10 day snapshot for each porosity ... a well connected icy porous matrix": the ice already spans the cell at $t = 0$ in 24 of 25 packings; what changes is that point contacts become necks. At the two highest porosities the network stays anisotropic.
   - [wording] "Next, we aim to better understand ... to better understand" (twice); "five different configurations for five separate porosities" (five packings at each of five porosities); "drop and roll" (hyphens); "gravitational affects" (effects); "a a".
   - [trim] The set-up (first half) moves to Section 2.5.
   - [add] What the colors show: vapor is undersaturated over convex ice, which sublimates, and supersaturated in necks and concavities, where it desublimates.

Figure 5 [written] placed. Caption: see "Figure captions" after Section 6.

4.2 Temperature Rescales Time

2. **Computing the conductivity** (L362–371)

   Applies the homogenization and describes the tensor.

   - [correct] "We obtain ... $\mathbf{K}_{\text{eff}}$ by integrating (17a) and applying the zero mean condition": it is the volume average of $\mathbf{K}(\nabla \mathbf{t} + \mathbf{I})$ over the cell (Eq. 10 in diffuse form).
   - [correct] "off diagonal components are at least one order of magnitude lower": k_{xy} reaches about 12% of k_{xx} on one packing; it is zero in the mean.
   - [wording] "changes ... is expected" (are); "figure figure 6"; "correspond the the"; "lower that" (than); "over various seeds" (packings).
   - [trim] The two sentences on solving Eq. (17) on the same mesh repeat Section 2.3.

- [add] Anisotropy: $k_{xx}/k_{yy} = 1.02,\ 1.03,\ 0.93,\ 0.82,\ 0.66$ from the lowest porosity to the highest, with y the direction of deposition. The scalar k_{eff} is used from here on.

3. **Figure 6, one packing at five temperatures** (L372–381)

   Conductivity and SSA against time and against sintering age.

   - [correct] "After the first day": that is 11 τ_{sub} at $-20\,^{\circ}\text{C}$ only; the relaxation period is a fixed number of τ_{sub}, so it is 0.26 d at $-5\,^{\circ}\text{C}$ and 7.7 d at $-40\,^{\circ}\text{C}$.
   - [correct] "the trend is linear": linear in $\log\theta$, about 0.13 per decade.
   - [wording] "local sintering occurs at high rates … and the phase-field interface relaxes" joins two different things; "non-dimensional" here, "nondimensional" in the caption.
   - [add] Panel (a) is not mentioned.
   - [add] The result as a number: against θ the rates follow τ_{sub} within 4%; against SSA the five temperatures agree within 0.27% for every one of the 25 packings (median 0.07%).
   - [add] A size that does not depend on ε: between $\theta = 30$ and 331, k_{eff} rises by a factor 1.171 ± 0.015 and SSA falls to 0.819 ± 0.007, for $\varphi \leq 0.375$.
   - [add] Why curves are normalized at $\theta_r = 30$ and not at $t = 0$: the first day depends on the interface parameter (Fig. S2).

   Cite: for SSA against log time in isothermal snow: Legagneux & Dominé 2005 `legagneuxMeanFieldModel2005`; Taillandier et al. 2007 `taillandierRateDecreaseSpecific2007` (functional form only; verify).

Figure 6 [written] placed. Caption: see "Figure captions" after Section 6.

4.3 Conductivity as a Function of Specific Surface Area

4. **Figure 7, porosity** (L382–400)

   One paragraph covering all four panels.

   - [correct] "15 unique packings–3 packings for each porosity": five packings at each of three porosities.
   - [correct] "We show the time evolution of k_{eff} for each of the simulations": panel (a) is $-20\,^{\circ}\text{C}$ only, the mean of five packings and their range.
   - [correct] "the ice matrix remains disconnected in the horizontal direction": not so. The ice spans the cell in both directions in every packing from 10 d on (percolation test). What happens is that the packings become anisotropic and the two directions follow different laws: along the deposition direction the exponent stays at -0.79 to -0.86 up to $\varphi = 0.425$; across it, -0.70, -0.60, -0.48 from $\varphi = 0.375$ on.
   - [correct] "The linear trend" in panel (c): the axis is logarithmic; the fit is exponential in porosity.
   - [wording] "may effect" (affect); "scientifically their connectivity as." is unfinished; "the strength of the reference value" (the reference value); "each of the the"; "A single point … show" (shows); "panels(c)".
   - [trim] One paragraph carries four panels; split into three: (a); (b); (c) and (d).
   - [add] Why four temperatures: the $-40\,^{\circ}\text{C}$ runs end at $\theta = 43$, too soon after the reference age.

- **[add]** The numbers. Exponent −0.80; by porosity −0.79, −0.82, −0.78, then −0.73, −0.63. Level $k_{\mathrm{eff},r}/k_{\mathrm{ice}} = 1.37\exp(-5.01\,\varphi)$ for $\varphi \leq 0.375$, 6.6% scatter between packings. At 30 d the conductivity differs by a factor of about 3.3 across porosity.
- **[add]** What it amounts to: SSA gives the relative change in conductivity; porosity gives the level; neither alone gives both.

Open: whether to plot or tabulate the directional exponents; whether to add that a porosity-dependent representative volume may contribute (not tested: the domain-size study was run at $\varphi = 0.325$ only).

Figure 7 **[written]** placed. Caption: see "Figure captions" after Section 6.

5. **What the scatter is** (optional; to write)

   Differences between packings in the exponent are properties of the packings, not noise.

   - **[add]** The scatter in the exponent does not fall with domain size while the scatter in the level does (11% to 4.5%).

   Open: include here, in Section 5.3, or leave out.

5 Discussion

Paragraphs 1–3 of Section 5.1 and the start of Section 5.2 are drafted.

(a) **Section opening** (L402–411)

Recaps the paper.

- **[correct]** "our model is able to capture the trends in sintering dynamics at multiple temperatures": at −20 °C it gives 41% and at −5 °C 170%.
- **[trim]** The third statement of the plan (after the Introduction and the opening of Section 3). Not needed.

(b) **Lineage of the model** (L412–424)

Kaempfer & Plapp, Moure & Fu, and what this paper adds.

- **[wording]** "They also, qualitatively validate"; "has since become popular method".
- **[trim]** Belongs in the Methods opening, in one or two sentences.

5.1 What Vapor Transport Alone Reproduces

1. **The unsaturated pair** (L426–452)

   Why the model gives 41% of Molaro's neck growth and what that does and does not mean.

   - **[correct]** "underpredicts … overpredicts … by 41% and 170%": the model gives 41% *of* the measured growth at −20 °C (it underpredicts by 59%) and 170% of it at −5 °C.
   - **[correct]** "It does not account for surface diffusion, which is expected [to] contribute significantly" (twice): by Molaro's own equations surface diffusion is below 1% of vapor transport at these neck sizes (Appendix A). Remove. Molaro et al. say so themselves (p.30: "Grains 100 μm in radius showed no significant effects from neglecting surface diffusion on any timescales"); their grains are 72–101 μm in radius. **[lit. check F3]**

Suggested text: At $T = -20\,^{\circ}\mathrm{C}$ our model gives 41% of the measured neck growth, and at $T = -5\,^{\circ}\mathrm{C}$ it gives 170%. Our model contains no surface diffusion, but by the equations and parameters of `molaroMicrostructuralEvolutionWater2019`, surface diffusion adds less than 1% to vapor transport at these neck sizes (`sec:Appendix_SD_VT`), so it does not explain the shortfall at $-20\,^{\circ}\mathrm{C}$.

- [**correct**] "This result was expected": it was not.
- [**correct**] "The chamber being unsaturated ... [is] likely the reason why our model predictions overestimate": the undersaturation is already in the model, through the inferred humidity.
- [**wording**] "preformed" (performed); "the the"; "our model own accounts" (only accounts); "expected contribute"; the three sentences from "Furthermore, our model also underestimates" appear twice.
- [**trim**] The first three sentences repeat Section 3.1.
- [**add**] "Even with the vapor diffusivity raised 30 or 100 times": say these were tests not shown, and that the neck was insensitive to the chamber humidity at the nominal diffusivity.
- [**add**] How small the driving force is: 4.8×10^{-4} falling to 9×10^{-5} in relative vapor density, the equivalent of 4.9 to 1.0 mK; the chamber's undersaturation is 2.85×10^{-3}, 6 to 31 times larger.
- [**add**] What that makes the result sensitive to, without choosing: (i) how much of the ambient undersaturation reaches the neck (about 6% in our run; with none, the one-equation neck model gives 66% of the growth and an exponent of 0.23, the value of our fit to Molaro's data); (ii) temperature differences of millikelvin between neck and grain surface; (iii) α_c above 0.1 for droplets quenched in liquid nitrogen.
- [**add**] Latent heat is not a candidate: in the run the neck is 0.002 mK warmer than the grain surface.
- [**add**] **The thin-interface term (user, 10-16: put its size here, and use it for the $-20\,^{\circ}\mathrm{C}$ shortfall).** At $\alpha_c = 0.1$, $\Delta\beta/\beta_{\mathrm{sub}} = 0.23$ at $-20\,^{\circ}\mathrm{C}$ (1.831×10^4 against $7.941 \times 10^4\ \mathrm{s\,m^{-1}}$, Table 4), against 0.02 in the aggregates and 0.002 in the saturated pair. The realized attachment is that of $\alpha_c \approx 0.081$, not 0.1, so it belongs in the explanation. **But it is a small part of it:**
  - At most it would raise the rate by 23%, and only if attachment alone limited growth. At $\alpha_c = 0.1$ gas diffusion carries much of the resistance at the neck ("above 0.1 the gain is small, because gas diffusion caps the rate", `studies/molaro_2019/envelope_ode/`).
  - The neck ODE over 78 min from the 14 μm start, with and without $\Delta\beta$ ($c = 0.275$): removing it adds 4.5% to the neck growth with the fitted undersaturation, 5.5% without it. That takes 41% to about 43% of the measured growth; the gap is 59 points. (Estimate: c was calibrated on our run, which already included $\Delta\beta$.)
  - The $-5\,^{\circ}\mathrm{C}$ pair has the same ratio (0.22) and overpredicts (170%), so the term cannot explain why $-20\,^{\circ}\mathrm{C}$ falls short.

  Say it in one sentence, as one reason the effective α_c sits below the nominal value, and keep the other candidates (above) as the main ones. A run at $\alpha_c = 0.123$ would realize 0.1 if you want the nominal value exactly.

Suggested text: At $\alpha_c = 0.1$ the thin-interface correction is 23% of β_{sub}, compared with 2% in the aggregate simulations, so our grain pairs attach vapor as if α_c were about 0.08. This slows neck growth by about 5%, because at this α_c diffusion of vapor through the air carries much of the resistance at the neck; it accounts for only a small part of the shortfall at $-20\,^{\circ}\mathrm{C}$.

- [add] Molaro's own model matches through a vapor term with $\alpha_c = 1$ and no gas diffusion, and they present its agreement as order-of-magnitude.
- [add] Why the saturated experiment is the better test: it removes the undersaturation.

Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`; Swinkels & Ashby 1981 `swinkelsSecondReportSintering1981`.

2. **The saturated pair, and exponents** (tex L502; **reframed 10-15**)

   The model reproduces how the exponent depends on saturation; the unsaturated value carries the starting-neck caveat.

   - [add] **Framing.** The model reproduces the dependence of the exponent on saturation that Demmenie et al. observed: 0.29 at saturation (theirs 0.26–0.33) and 0.135 in the undersaturated $-20\,^{\circ}\mathrm{C}$ case (theirs about 1/7 when the ice sublimated). The draft's "they are lower than measured values" no longer holds at saturation.
   - [correct] "As two spherical ice grains sinter in an environment with a saturated water vapor field, the growth is expected to be kinetics limited": saturation does not make growth attachment-limited; a small α_c does. The $t^{1/3}$ law assumes it.
   - [add] **Why 1/3 at small α_c** (replaces the sentence above; reasoning in Section 3.2 and Appendix B). Two sentences:

     Suggested text: An exponent near 1/3 is expected only when attachment at the ice surface, rather than diffusion of vapor through the air, limits the growth of the neck: the resistance to attachment does not depend on the size of the neck, whereas the diffusion path lengthens with the radius of curvature of the neck, which lowers the exponent to 1/5 (`kingery_regelation_1960, Hobbs_and_Mason, blackfordSinteringMicrostructureIce2007`). With $\alpha_c = 10^{-3}$, diffusion through the air contributes less than 2% of the resistance at the neck, so our saturated case lies in the regime the $t^{1/3}$ law describes.

   - [correct] Numbers: "$a = 0.21$ across the entire simulation ... as high as $a = 0.236$" describe the pre-necked run: $a = 0.29$, 0.277 to 0.289 as the window moves.
   - [correct] [lit. check F4] "a much better match for the theorized scaling ... $t^{1/7}$" (tex L502): 1/7 is not a theory of unsaturated sintering. It is the classical surface-diffusion exponent (Kingery 1960, Table I) and the value Demmenie et al. measured when "we deliberately set the humidity too low and let the ice sublimate" (p.3). Our model reaches it with vapor transport alone. Add that this run starts from a pre-made neck (Section 3.2, open item).
   - [add] "We additionally compare our measured interface velocity ..." states no result. The numbers in Appendix B (2.5 times the Gibbs–Thomson value at 2 h, 1.0 by 71 h) are for the pre-necked run; **recompute them for the tangent-start run** (`interface_width_estimate.py`) or drop the sentence.
   - [add] Why not surface diffusion: mechanisms in parallel add their rates, and surface diffusion alone gives $t^{1/7}$, so it could not raise an exponent; at saturation none is needed.
   - [wording] The first sentence is unfinished ("growing under as ."); "where observe"; "we had considered"; "$1.002e-03$"; "preform" (perform); two sentences in a row begin "We additionally" and "Additionally".

Suggested text: At saturation, the grains neither grow nor shrink and the neck width follows $w \propto t^{0.29}$ (Section 3.2), within the range that `demmenieIceSinteringSublimation2025` measured, 0.26–0.33. In the undersaturated case at $T = -20°C$, where the grains sublimate, we measure $a = 0.135$, close to the exponent of about 1/7 that `demmenieIceSinteringSublimation2025` measured when the humidity was too low and the ice sublimated. The same value is the classical exponent for surface diffusion (`kingery_regelation_1960`), which our model does not contain; here it arises from vapor transport alone. Our model therefore reproduces how the exponent depends on the saturation of the vapor.

Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`; Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`; Kingery 1960 `kingery_regelation_1960`.

3. α_c **as an effective parameter** (L469–477)

What the uncertain coefficient does and does not change.

- **[correct]** "do not change how the microstructure evolves, just the rate": true only while growth is limited by attachment. α_c sets the time scale ($\tau_{\text{sub}} \propto 1/\alpha_c$) *and* the group $\Lambda = L^*/\bar{R} \propto 1/\alpha_c$ that decides how much gas diffusion matters. In the aggregates $\Lambda \approx 2.8$. A smaller α_c only rescales time; a larger one brings diffusion in and changes the path as well. (An earlier version of this outline said a tenfold larger α_c would leave the curves unchanged against θ; that was wrong.)
- **[correct]** "$\Delta\beta/\beta_{\text{sub}} = 0.02$" for both the aggregates and the saturated pair: it is 0.02, 0.002, and 0.23 for the Molaro pairs.
- **[correct]** **[lit. check F7]** The sentence on Bouvet et al. rescaling time through α_c (tex L504) cites the 2023 paper, a laboratory experiment with no model. It is Bouvet et al. (2022, p.2: "Relating the numerical time … to the real physical time involves the condensation coefficient"). Their model is limited by attachment alone, so the rescaling is exact there; in ours it holds while attachment limits growth (previous bullet).

  Suggested text: `bouvetSnowEquiTemperatureMetamorphism2022` used this to calibrate their model: matching the SSA evolution measured by μCT fixed α_c, and with it the physical time of their simulations. In their model growth is limited by attachment alone, so the rescaling is exact; in ours it holds while attachment, not vapor diffusion, limits growth.
- **[wording]** "The kinetic attachment parameter, and thus condensation coefficient, do not"; "μm-CT" (μCT).
- **[trim]** Eq. (τ_{sub}) and the sentence after it move to Section 2.2.
- **[add]** What α_c does to the two grain-pair comparisons: Molaro's rate wants a large value (41% at 0.1; 51–75% at 0.8 in the one-equation neck model), while the sharp-interface exponent at saturation falls with it (0.29 at 10^{-3}, 0.25 at 10^{-2}, 0.18 at 0.1). One value cannot serve both.
- **[add]** **[from the notes]** Bouvet et al. fit $\alpha_c = (9.8 \pm 0.7) \times 10^{-4}$ to tomography at $-2\,°C$; Braun et al. find 10^{-3}–10^{-1}, changing as surfaces facet, and call it an effective, surface-averaged value.

Cite: Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Braun et al. 2024 `braunRigorousApproachSpecific2024`; Granger et al. 2021 `grangerOrientationSelectiveGrain2021` for faceting under a gradient; Libbrecht & Rickerby 2013 `libbrechtMeasurementsSurfaceAttachment2013` as background on why α_c is uncertain.

5.2 Temperature as a Rescaling of Time

4. **Why the curves superpose** (L479–487; breaks off at "Since we have already")

   A similarity argument: in dimensionless form, almost all of the temperature dependence is in one time scale.

   - **[correct]** "β_{sub}, which is a free variable": it is set by α_c, which is chosen, and it depends on temperature through $\rho_{vs}^{I}(T)$. That dependence is the point of the paragraph.
   - **[wording]** "is evident … and is evident"; "a two part series" (two steps in series); "drawn to areas of lower vapor saturation".
   - **[add]** Step 1. Scale lengths by $\bar{R}$, supersaturation by $d_0/\bar{R}$, time by $\tau_{\bar{R}} = \bar{R}^2\beta/d_0$ ($= 2500\,\tau_{\text{sub}}$). The interface condition becomes parameter-free. The saturation vapor density, with its 31-fold change from -40 to $-5\,^{\circ}\text{C}$, is entirely inside the time scale.
   - **[add]** Step 2. One temperature-dependent group is left, $\Lambda = L^*/\bar{R}$ with $L^* = (D_v/\alpha_c)\sqrt{2\pi m/k_B T}$. The saturation vapor density cancels in it; what remains is the diffusivity. It changes by 20% where the time scale changes by a factor of 29.
   - **[add]** Step 3. So the saturation vapor density sets the clock, and the diffusivity is a weak remaining correction: the likely source of the 4% by which rates depart from τ_{sub} (consistent in size; not tested as the cause).
   - **[add]** Say plainly that with one sintering mechanism this is close to a consequence of the construction; the numerical result is how well it holds. Do not say gas diffusion is negligible ($\Lambda \approx 2.8$); what is small is its change with temperature.
   - **[add]** The two-page derivation (`temperature_scaling/time_temperature_scaling.pdf`) can be an appendix or supporting information.

5. **What would break it, and the activation energy.** (new) **[to write]**

   Message: *The collapse is therefore a signature of a single route. Its temperature dependence differs from the one measured for strength gain, which points to a second route in real samples and gives experimenters something to measure.*

   The clock's temperature dependence is an activation energy of about 50 kJ mol^{-1}, that of the saturation vapor density. **[from the notes]** Choukroun et al. measure 24.3 ± 3.3 kJ mol^{-1} for strength gain and attribute it to surface self-diffusion (L613–620), at 193–243 K; the two temperature ranges overlap at 233–243 K. The difference is consistent with a second route in their samples, and an activation energy is something a laboratory can measure to tell the routes apart. A temperature-dependent α_c, a second route, or a temperature gradient would each break the superposition, which makes it a test.

   Cite: Choukroun et al. 2020 `choukroun_strength_2020`.

5.3 Specific Surface Area as the State Variable

6. **Density is not enough.** (new) **[to write]**

   Message: *Conductivity changes while porosity stays fixed, so a density-based formula cannot describe a sintering deposit.*

   In these runs porosity is fixed and conductivity still rises, by 17% between $\theta = 30$ and 331. **[from the notes]** Calonne et al. (2011): tensor from 30 tomographic images, a regression on density, scatter attributed to anisotropy (L507–517). Calonne et al. (2014): under a temperature

gradient conductivity changes at constant density, with SSA and anisotropy (L517–529). Half the length of the notes.

Cite: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`; Calonne et al. 2014 `calonneStudyTemperatureGradient2014`; Schneebeli & Sokratov 2004 `schneebeliTomographyTemperatureGradient2004` (the notes cite it for the same behavior under a gradient).

7. **The power law and where it stops.** (new) **[to write]**

   Message: *State the law, its range, and that outside the range we make no claim.*

   $p = -0.80$ for $\varphi \leq 0.375$; the level from porosity; above 0.375 the exponent weakens and the packings are anisotropic. No claim that packings of different porosity converge.

   Cite: Calonne et al. 2011, 2019 `calonneNumericalExperimentalInvestigations2011`, `calonneThermalConductivitySnow2019` for conductivity against density, as the contrast.

8. **Why packings differ.** (new) **[to write]**

   Message: *Packings of the same porosity conduct differently because their ice networks are arranged differently; at high porosity the networks are anisotropic and the simple law fails in one direction.*

   Packings of one porosity differ in how well the ice is connected, and the exponent follows that (correlation -0.77 with conduction efficiency), not coordination number or anisotropy. In these packings the ice always spans the cell and the pore never does (percolation test, 2026-10-12): the pore space is a set of closed pockets, which is what "sealed pores" means here, and in two dimensions only one phase can span. What differs between packings is the quality of the ice network, not whether it exists: at high porosity the packings are anisotropic, grains dropped under gravity being supported from below with fewer contacts sideways, and the power law survives along the deposition direction while failing across it. Absolute values are therefore not compared with three-dimensional measurements, where both phases are connected. *Open: literature thresholds for discs are not yet checked against a source; whether to show the efficiency; the Euler-characteristic series (tested on three runs) is available if a direct measure is wanted.*

5.4 What a Laboratory Could Measure

9. **Three tests.** (new) **[to write]**

   Message: *Turn the results into measurements someone could make on sintered ice-sphere samples.*

   On unloaded ice-sphere samples held isothermally: (i) curves at different temperatures superpose under a time shift, and the shift gives an activation energy; (ii) conductivity against SSA is one curve; (iii) SSA falls linearly in log time. **[from the notes]** SSA is measured by near-infrared reflectance, gas adsorption and tomography (L562–568; Gallet, Matzl, Proksch).

   Cite: Gallet et al. 2009 `galletMeasurementSpecificSurface2009`; Matzl & Schneebeli 2006 `matzlMeasuringSpecificSurface2006`; Proksch et al. 2015 `prokschDensitySpecificSurface2015`; for isothermal SSA decay in the laboratory, Legagneux & Dominé 2005, Taillandier et al. 2007, Kaempfer & Schneebeli 2007 `kaempferObservationIsothermalMetamorphism2007`, Löwe et al. 2011 `loweInterfacialStructuralRelaxations2011` (forms only).

5.5 Implications for Plume Deposits on Enceladus

10. **What carries over.** (new) **[to write]**

Message: *For Enceladus the qualitative conclusion holds: a sintering deposit conducts better as its surface area falls, even with no compaction.*

Qualitative: as a deposit sinters, SSA falls and conductivity rises together, without densification. **[from the notes]** Tiger Stripe temperatures up to about 180 K (Spencer); grain size increasing toward the fractures in VIMS data (Jaumann; L580–602); Molaro et al. on the dependence of sintering on temperature and grain size.

Cite: Spencer et al. 2009 `spencerEnceladusActiveCryovolcanic2009`; Spencer & Nimmo 2013 `spencerEnceladusActiveIce2013`; Jaumann et al. 2008 `jaumannDistributionIcyParticles2008`; Molaro et al. 2019; Choukroun et al. 2020.

11. **Why it matters, and the gap.** (new) **[to write]**

Message: *Conductivity of the surface layer affects thermal and radar interpretation, but our conditions are far from the moon's, so we give no timescales.*

[from the notes] A low-conductivity surface layer changes subsurface temperatures and radar attenuation (Byrne; L603–612); Choukroun et al. on strength gain and fallout rates of 0.01–1 $\mathrm{mm\,yr^{-1}}$ (L613–629). Then the gap, stated plainly: the runs stop at $-40\,^{\circ}\mathrm{C}$ and use 50 μm grains in air; the moon is far colder, the grains are micron-sized, and there is no air. No timescales are given.

Cite: Byrne et al. 2024 `byrneRadarAttenuationSubsurface2024` (a preprint DOI in the `.bib`; check whether it is published); Choukroun et al. 2020; possibly Ferrari et al. 2021 `ferrariThermalEmissionSaturns2021` on thermal emission of Saturn's icy moons (in your `.bib`; check relevance).

5.6 Limitations

12. **Model.** (new) **[to write]**

Message: *What physics is missing.*

Vapor route only; constant α_c; air-filled, sealed pores at 1 atm; conduction only and constant k_{ice}, k_{air}. **[from the notes]** Fourteau et al. on the contribution of vapor transport to apparent conductivity under a gradient (L569–575; check the figure quoted); Calonne et al. (2019) on temperature-dependent conductivities (L575–579).

Cite: Fourteau et al. 2021 — your `.bib` has two, `fourteauImpactWaterVapor2021` and `fourteauMacroscopicWaterVapor2021`; the point about apparent conductivity is the first. Calonne et al. 2019 `calonneThermalConductivitySnow2019`.

13. **Numerics and geometry.** (new) **[to write]**

Message: *What the simulations cannot resolve or represent.*

Two dimensions, so relative changes are claimed and absolute conductivities are not; one grain size distribution; rounded grains (discs) in point contact, so the rounding of fresh, dendritic snow, the fast first stage of isothermal metamorphism (Kaempfer & Schneebeli 2007, p.1, p.5), is outside the model; the first day depends on the interface parameter; the Molaro runs start from a neck put in by hand, which biased the fitted exponent low in the saturated case (0.21 against 0.29 from contact; Appendix B); the Molaro runs use time steps of 22 and 49 τ_{sub}, outside the range the time-step study covered; pore throats narrower than the diffuse band (9.2 μm) are not resolved, which raises k_{eff} most at low porosity; the power law is established for $\varphi \leq 0.375$; rates scale with $1/\alpha_c$, whose value is uncertain over two decades.

Cite: none needed beyond those above.

In the notes and not to be carried over: $\beta_{\mathrm{sub}}(T)$ from Libbrecht's $\sigma_0(T)$, a single tuning and a "genuine prediction" at $-5\,^{\circ}\mathrm{C}$ (L454–481); isotropy, and a collapse across porosity (L530–561); sintering "on the order of decades" (L596) and conductivity starting "near 0.0" (L611).

6 Conclusions

1. **What was done and found.** (new) [to write]

   Message: *Three results, each with the range over which it was shown.*

   One sentence of method, then the three results with their ranges: temperature rescales time (-40 to $-5\,^{\circ}\mathrm{C}$, within 0.3%); conductivity follows $\mathrm{SSA}^{-0.80}$ with the level set by porosity ($\varphi \leq 0.375$); vapor transport gives 41% of measured neck growth at $-20\,^{\circ}\mathrm{C}$.

2. **What comes next.** (new) [to write]

   Message: *The experiment and the model extensions that would test or extend them.*

   The laboratory test of Section 5.4; a second transport route in the model; three dimensions; colder temperatures and smaller grains.

Figure captions

Checked against the figure files in the 10-14 upload (rendered here), not only against the text. AGU asks that a caption let the figure be read on its own: what is shown, the conditions, what each line, symbol and color means. Four points apply to several figures:

- [correct] **The σ color scale** (Figures 3–6) is relative to a flat ice surface and is clipped. Say both, and give the range. Without this a reader cannot tell why the pore is dark red at saturation (Figure 4) or uniformly dark blue in Molaro's chamber (Figure 3). The scale is symmetric-logarithmic (ticks at ± 0.01 and $\pm$ the maximum): say so.
- [wording] **"vapor saturation filed is colored according the the blue/red colorbar"** (Figures 3, 4): "the supersaturation σ is shown on the blue–red scale". "pale-blue" and "pale blue" both occur: one form.
- [wording] **The phase-field colorbar is labeled** ϕ_i in the art of Figures 3, 4, 6 and in the Figure 2 caption; the text writes ϕ. One symbol (decided 09-28: ϕ).
- [add] **Which run.** Every image caption should name the porosity, temperature and domain of what is shown; most do not.

1. **Figure 1, mechanisms and an aggregate** (L268)

   (a) a grain pair before and after necking; (b) an aggregate at four times.

   - [correct] **The art says $R(t)$, the caption $w(t)$.** The arrow runs from the axis to the neck surface, i.e. a neck *radius*, while w is the neck *width* everywhere else (Figures 3, 4; Molaro's convention). Relabel the arrow $w(t)$ and draw it across the whole neck, or label it $w/2$.
   - [written] The caption is vapor-only (10-15). [wording] "Ice growth as a result fo vapor transport is shown": "of"; or "Arrows show vapor transport". It still says "a neck of radius $w(t)$" (next bullet).

- [add] (b): which run (porosity, temperature, seed) and that the panels are at 0, 10, 19 and 30 days; the scale bar is 100 μm (it is a crop, so say "a region of").
- [wording] "A 2-D cross-section of two spherical ice grains that have undergone sintering. The initially non-sintered touching ice grains (shown as blue dotted line)": "Two ice grains in cross-section, before (dotted) and after (solid) a neck of width $w(t)$ has grown". The grey crosshair lines in (a) are unexplained: remove, or say they are the axes of symmetry.

2. **Figure 2, the homogenization** (L(after 2.3))

   (a) a cell, (b) the interface and mesh, (c) the corrector, (d) the heat flux.

   - [add] Not cited in the text (Section 2.3 is the place).
   - [add] Which sample: porosity, temperature, age, and that the cell is 2 mm wide (the scale bar is 500 μm).
   - [correct] (b): the dashed lines are orange, not red, and are the $\phi = 0.05$ and 0.95 contours; the solid line is $\phi = 0.5$. Write ϕ, not ϕ_i.
   - [add] (c): the black contours are the ice–air interface; only the x cell problem is shown ($\mathbf{t}$ has a second component from the y problem). Missing period after (c).
   - [add] (d): color is on a logarithmic scale; "per unit mean gradient in the x direction" is right; "brightness of the path": "brighter paths carry more heat".
   - [wording] "the mesh in which we solve our homogenization model over": "the mesh on which the cell problem is solved".

3. **Figure 3, Molaro et al.** (L(Section 3.1))

   (a) the pair at the start and at 78 min, (b) neck width, (c), (d) grain diameters.

   - [correct] **The color scale does not show the chamber.** The far field is undersaturated by $\sigma \approx -2.9 \times 10^{-3}$ (99.715% relative to flat ice), a hundred times beyond the $\pm 0.28 \times 10^{-4}$ scale, so the whole pore sits at the bottom color. Say the scale is clipped to resolve the neck, and give the far-field value.
   - [correct] "On the left is the initial configuration": it is the first measured frame ($t = 0$ is Molaro's first measurement, and the simulation starts from the measured neck). Say so; it is why the curves in (b) start on the first data point. "78 minutes": check against Molaro's text, which describes their Figure 11 as 57 minutes apart (Section 3.1 entry).
   - [add] (b): both temperatures are shown ($-20\,^\circ$C blue, $-5\,^\circ$C vermilion); lines are the model, circles the data, error bars are Molaro's stated uncertainty (check what they are); the numbered circles mark the frames in (a). Give the grain diameters.
   - [correct] (c), (d): "the experimentally measured grain size through time relative to the large and small grains": they show the diameter of the large (c) and small (d) grain divided by its value at $t = 0$; solid lines are the model, circles the data, dashed lines linear fits to the data.
   - [trim] "The simulation is designed to replicate the conditions and geometry ...": belongs in Section 3.1; the caption only needs the conditions.

4. **Figure 4, the saturated pair** (L(Section 3.2); **rebuilt 10-15** from the tangent-start run)

   (a) the pair at 0 and 150 h, (b) neck width with the fitted law ($a = 0.29$) and the 1/3 law.

   - [correct] The caption describes the old figure: 100 h → 150 h; the $t = 0$ frame now shows two grains just touching.

- [add] **Why the pore is red.** σ is relative to flat ice, and Section 3.2 sets the far field to the Gibbs–Thomson value of the grains, $2d_0/R \approx 2.4 \times 10^{-5}$ for $R = 84$ μm, the top of the scale. Say this; a reader expects a saturated pore to be white.
- [add] The legend: $C(t + t_0)^a$, $a = 0.29$, and $C(t + t_0)^{1/3}$. Say both are fitted (t_0 free), over 20 min to 150 h; temperature $-20\,^\circ$C, $R = 84.4$ μm, $\alpha_c = 10^{-3}$; the numbered circles mark the frames in (a). Cite Demmenie et al. for the measured range and Appendix B for the 0.29 prediction.
- [wording] "An isolated ice grain pair shown undergoing sintering shown": "Two equal ice grains sintering in saturated air, at $t = 0$ (left) and 150 h (right)". "filed", "the the".

5. **Figure 5, the gallery** (L(Section 4.1))

   One packing at each of five porosities, at four times, at $-20\,^\circ$C.

   - [wording] "Five microstructures of varying porosity φ coarsening due to DSM": "One packing at each of five porosities, sintering at $-20\,^\circ$C". Say which seed, and that each cell is 2 mm wide.
   - [add] The vapor colorbar has words but no numbers: give the range, and that σ is relative to flat ice and clipped, as in Figure 6.
   - [add] The times (0, 10, 19, 30 d) are the dashed lines in Figure 7(a); one clause linking them.

6. **Figure 6, one packing at five temperatures** (L(Section 4.2))

   (a) four frames, (b) conductivity against time, (c) against sintering age, (d) SSA against sintering age.

   - [correct] (a): "$|\sigma| \leq 0.31$": the colorbar says 0.34×10^{-4}. The factor 10^{-4} is missing from the caption.
   - [add] Which packing (it appears to be the $\varphi = 0.325$ packing of Figure 5; say so) and that the frames are at $-20\,^\circ$C, at 0, 10.2, 19.5 and 30.0 days.
   - [add] Define $\theta = t/\tau_{\text{sub}}$ and $\theta_r = 30$ (the dotted line). Say why the curves in (b) start at different values: each is normalized at θ_r, which falls at a different time at each temperature.
   - [add] (c), (d): the five curves overlie (within 0.3%); that is the result, and the caption can say it. Axes start at $\theta = 1$ (log scale).
   - [wording] "preformed" (performed); "2.0mm" (2.0 mm); "The vertical axis is k_{eff}, average value of k_{xx} and k_{yy}, is normalized": "$k_{\text{eff}} = (k_{xx} + k_{yy})/2$, normalized ...". "Shows the same curves plotted for": "(c) The same curves against".

7. **Figure 7, porosity** (L(Section 4.3))

   (a) conductivity against time, (b) against SSA, (c) its reference level, (d) its exponent.

   - [correct] (a): "The solid line corresponds to a single seed and the highlighted areas are the seed scatter": the line is the mean of five packings and the band their minimum–maximum (as `compare_keff.py` draws it). These are 2-D conductivities in $\mathrm{W\,m^{-1}\,K^{-1}}$; say they are not those of 3-D snow (decided 09-23: no absolute value is claimed).
   - [correct] (b): only $\varphi \leq 0.375$ is shown (three colors); the caption speaks of "the porosities in panel (a)". Say the higher porosities are left out and why (Section 4.3). $-40\,^\circ$C is absent because those runs end at $\theta = 43$; say so. The grey arrow is the direction of time.

- [correct] (c), (d): one point per packing, temperatures pooled (they coincide at θ_r), not "each of the simulation runs". The dashed line in (c) is a fit to $\varphi \leq 0.375$, extended; give its form. The grey band marks the porosities left out of the fits.
- [wording] "Plots the the"; "Shows" at the start of (c), (d); "15 unique packings were used to model DSM at temperatures": "15 packings at each of −30, −20, −10 and −5 °C (60 curves)".

Open Research; Acknowledgments

[to write] Code and data availability; run inventory (`RUN_TABLE.md`). Acknowledgments.

Supporting Information

- **Figures.** S1 domain-size convergence; S2 interface parameter; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Tables, text.** Complete parameters, if moved; packing construction; the weak-form derivation, if trimmed; run table.

Appendix: the two calculations behind Sections 3 and 5.1

A. Surface diffusion against vapor transport, by Molaro's own equations

Question. Does surface diffusion supply the 59% of neck growth our model lacks at −20 °C?

Equations (Molaro et al., 2019, Table 3 and Eqs. 1–4, 6, 17–20; a grain radius, x neck radius, S their smoothing factor):

$$\dot{V}_1 = S\,\frac{3\pi x\,\delta_s D_s\,\gamma_s\Omega\,(K_3 - K_2)}{d_2\,kT} \qquad \text{surface diffusion}$$

$$\dot{V}_3 = S\,2\pi x\rho\theta\,P_v\,\frac{\gamma_s\Omega}{kT}\left[\frac{\Omega}{2\pi\Delta_o kT}\right]^{1/2}(K_3 - K_m) \qquad \text{vapor transport}$$

$$\dot{x} = \frac{1}{2\pi x\theta\rho}\sum_i \dot{V}_i$$

$$K_3 = \frac{2}{a},\quad K_m = -\frac{1}{\rho} + \frac{1}{x},\quad \rho = \frac{x^2}{2(a-x)},\quad \theta = \tan^{-1}\frac{a}{x+\rho},\quad K_1K_2 = K_m^2,$$

$$\frac{K_2 - K_1}{K_1} = \frac{3A\rho}{x} + \frac{3}{2} - \frac{3}{2}\left[3\left(\frac{A\rho}{x}\right)^2 + \frac{4A\rho}{x} + 1\right]^{1/2},\quad d_2 = \frac{\rho}{1 + \left(\frac{4(K_2-K_1)}{3K_2}\right)^{1/2}},\quad A = \frac{\delta_b D_b}{\delta_s D_s}.$$

The vapor term is Hertz–Knudsen with $\alpha = 1$ and no gas diffusion (their Appendix A).

Parameters (their Table 5): $\Omega = 1.181\times10^{-29}$ m^3; $\Delta_o = 917$ kg m^{-3}; $\gamma_s = 0.06$ J m^{-2}; $\delta_s = \delta_b = 9.04\times10^{-10}$ m; $D_{0s} = 1.4\times10^{-8}$ m^2 s^{-1} (literature range down to 1.35×10^{-13}), $Q_s = 23.2$ kJ mol^{-1}; $D_{0b} = 8.4\times10^{-4}$ m^2 s^{-1}, $Q_b = 4.9$ kJ mol^{-1}; P_v from Murphy & Koop (2005), 103 Pa at −20 °C.

At their experiment ($a = 87$ μm, −20 °C; measured mean $\dot{x} = 3.4\times10^{-9}$ m s^{-1}):

x [μm]	ρ [μm]	$\dot{V}_1/\dot{V}_3$	$\dot{x}$ from $\dot{V}_1$ [m/s]	$\dot{x}$ from $\dot{V}_3$ [m/s]	$\dot{V}_1/\dot{V}_3$, $A=0$
16.4	1.9	6.7×10^{-3}	5×10^{-11}	7.8×10^{-9}	8×10^{-7}
20.0	3.0	3.1×10^{-3}	2×10^{-11}	4.9×10^{-9}	3×10^{-7}
25.0	5.0	1.3×10^{-3}	4×10^{-12}	2.9×10^{-9}	1×10^{-7}
32.4	9.6	4.1×10^{-4}	6×10^{-13}	1.5×10^{-9}	4×10^{-8}

- Surface diffusion is 0.7% falling to 0.04% of their vapor term. It exceeds it only for x below about 4 μm.
- Their vapor term alone gives the measured rate.
- The last column switches off the coupling to grain-boundary diffusion ($K_2 = K_m$). With their parameters $A = 3.5\times10^8$, and the coupling raises the driving curvature K_2 by a factor of about 4000.

Check and its limit. Their Figure 10 (25 μm, $T/T_m = 0.8$) was recomputed. The vapor current agrees (2×10^{-21} rising to 3×10^{-20} m^3 s^{-1}; theirs about 10^{-21} to 2×10^{-20}). The surface current starts about 100 times above theirs, because they solve the four coupled equations iteratively where this uses the approximate Eq. (20). The table therefore *overstates* surface diffusion. Script and output: `studies/molaro_2019/sa81_mechanisms/`.

What it implies. The gap is between our vapor route ($\alpha_c = 0.1$, diffusion through air) and whatever supplies vapor in the experiment. A one-equation neck model fitted to the same data wants $\alpha_c \approx 0.3$–1 (`studies/molaro_2019/envelope_ode/`).

B. The saturated pair against the Gibbs–Thomson law

Result (10-15). From grains just touching, the run gives $a = 0.289$, on the prediction below (0.29). The tables that follow are for the earlier *pre-necked* run, kept as the record of why the initial condition matters. **Question it answered:** was that run's 0.21 physics or resolution? Neither: it was the starting neck.

Prediction. For a neck of radius x between grains of radius R, with fillet radius $\rho = x^2/(2(R-x))$, fed from the grain surface through attachment in series with gas diffusion over a length $c\rho$:

$$v_n = \frac{d_0\left(\dfrac{1}{\rho} - \dfrac{1}{x} + \dfrac{2}{R}\right)}{\beta_{\mathrm{sub}} + c\,\rho\,k_D}, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\pi}{2}\,v_n, \qquad k_D = \frac{\rho_{\mathrm{ice}}}{D_v\,\rho_{vs}^I}.$$

Attachment alone gives $x^3 \propto t$; diffusion alone $x^5 \propto t$. Why: with attachment limiting, $v_n \approx d_0/(\beta_{\mathrm{sub}}\rho)$ and $\rho \approx x^2/2R$, so $dx/dt \propto R/x^2$; with diffusion limiting, the extra factor $1/(c\rho k_D)$ gives $dx/dt \propto R^2/x^4$. The gas resistance grows with the neck; the attachment resistance does not. This is Kingery's evaporation–condensation law ($n = 3$) against Hobbs & Mason's diffusion through air ($n = 5$; Blackford 2007, p.16). $R = 84.4$ μm, $-20\,^\circ$C, $\alpha_c = 10^{-3}$; $c = 0.275$ from the Molaro run, not refitted. Here gas diffusion is 0.4–2% of the resistance. Integrated from the run's first sample after relaxation, it gives a final width of 60.6 μm (run: 63.2) and an exponent of 0.29 (run: 0.21).

Run against prediction ($\varepsilon = 0.12$ μm; one interface band $= 9.19\,\varepsilon = 1.1$ μm; exponents from fits of $C(t+t_0)^a$ starting at the time shown):

Start	w [μm]	w/ε	fillet [bands]	velocity, run / prediction	a, run	a, prediction
2 h	35.1	293	2.1	2.54	0.215	0.292
6 h	39.2	327	2.7	1.72	0.221	0.292
16 h	44.9	374	3.7	1.36	0.227	0.290
31 h	50.1	418	4.8	1.18	0.231	0.288
51 h	55.0	459	6.0	1.08	0.234	0.287
71 h	58.8	490	7.1	1.02	0.236	0.286

The excess fits $1 + 11.4\,(\text{band}/\text{fillet})^{2.7}$ (rms 0.035).

Sensitivity to the assumed contact time (same data, t_0 fixed):

t_0	4.8 h (free fit)	8 h	12 h	16 h	24 h
exponent	0.212	0.234	0.258	0.281	0.325
rms misfit	0.28%	0.78%	1.33%	1.73%	2.31%

The prediction takes 16 h to grow a neck to the run's first fitted width.

What it shows and what it does not. The rate converges on the sharp-interface law, and the early excess accounts for most of the difference in exponent. Two causes fit these numbers equally: the interface being too wide for a small fillet, or the pre-made neck relaxing from its imposed shape. The August run from tangent contact, at twice this interface width, showed no excess and gave 0.283, which points to the pre-made neck. The run from contact at this interface width (10-15) settles it: $a = 0.289$, so the low exponent came from the pre-made neck. From a pre-made neck the exponent does not reach the prediction within 100 h (0.236 at the end against 0.29); from contact it does at once. Still to do: the velocity-over-prediction table for the tangent-start run. Scripts and output: `studies/molaro_2019/demmenie/` (`predict_neck_growth.py`, `interface_width_estimate.py`).

C. What the appendices of the manuscript should hold

The draft now has two appendix headings with these titles. Suggested content, each under a page:

- *Appendix A, Surface diffusion against vapor transport:* the two rate equations and the curvature relations, the parameter list, the four-row table, and the two sentences on what was and was not reproduced of Molaro's Figure 10.
- *Appendix B, The saturated pair against the Gibbs–Thomson law:* the neck law, the run-against-prediction table, the contact-time table, led by the tangent-start run against the prediction (0.289 against 0.29), then one paragraph and the contact-time table showing what a pre-made neck does to a fitted exponent.
- The time–temperature scaling (two pages, in `temperature_scaling/`) is a candidate third appendix, or supporting information, for Section 5.2.

Citations: what the .bib lacks

Suggested above and present in the `.bib`: all of them. Mentioned in this plan and *not* in the `.bib`:

- Auriault, Boutin & Geindreau (2009), the book whose naming of **t** the paper follows and which Calonne et al. (2014, p.6) cite for the cell problem (doi:10.1002/9780470612033; PDF in the library).
- Ettrich et al. (2014), the tensor law applied to heat conduction; only if a second reference for the law is wanted.
- A source for packing and percolation thresholds of discs, if Section 5.3 states them.

[written] Since 10-15 the `.bib` has Hobbs & Mason (1964), Kingery (1960) and Thomas et al. (1994); Kuczynski and Kingery & Berg are no longer needed, since Kingery (1960, Table I) and Demmenie et al. (Table 1) give the full set (F4). Blackford (2007) and Maeno & Ebinuma (1983) are in it too. Still to fix in the `.bib` (F11): Byrne et al. (2024) is the preprint; `noauthor_sintering_nodate` duplicates `Hobbs_and_Mason` and `noauthor_strength_nodate` duplicates `choukroun_strength_2020`; old-style keys `choukroun_strength_2020`, `nicoli_tensorial_2011`, `spitale_association_2007`, `postberg_sodium_2009`, `postberg_macromolecular_2018`, `kingery_regelation_1960`, `thomas_crushing_1994`, `Hobbs_and_Mason`.

Corrections that recur throughout

Paragraph-specific corrections are under each paragraph. These apply everywhere:

- **Notation.** Subscripts "ice" and "air" in place of i and a (ρ_{ice}, ϕ_{ice}, ϕ_{air}); $\mathbf{K}_{\text{eff}}$ for the tensor and k_{eff} for its scalar mean; k_{ice}, k_{air} as in the tables (the text has K_{ice}); ρ_{air}, P_{atm} in Eq. (2); ϕ against ϕ_i for the same field in Section 2.3.
- **Temperature.** T in °C (user, 10-16), with $T + T_0$ ($T_0 = 273.15$ K) only in Eqs. (3), (5), (7) and d_0; $T_0 = 273.15$ K only; the tables give the run temperature as T.
- ε is the interface parameter, not the interface width: fixed in Section 2.1 ("controls the width"); one "interface width" remains (Section 2.3).
- **"effect"** for "affect" (six places); "preform" for "perform" (four, incl. "preformed" in ¶4); doubled words where `\cref` already prints the word ("table table", "figure figure", "section section").
- α_c written 1.002×10^{-3}, not "1.002e-03".
- **Not cited in the text:** Table 2, Table 3, Figure 2.